Hybrid Gains and Tasty Grains: Comparing Production- and Consumption-Focused Extension to Accelerate Maize Varietal Turnover in Uganda

Bjorn Van Campenhout[*], Leocardia Nabwire[†], Berber Kramer[‡],
Carly Trachtman[§], Gashaw T. Abate[§]

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Abstract

Accelerating varietal turnover, replacing older crop varieties with newer, higher-yielding and more resilient ones, is a central goal of seed-system policy in sub-Saharan Africa, yet how best to promote it remains unclear. Because smallholders are both producers and consumers, extension can emphasize a variety's agronomic performance or its consumption qualities. We compare two instruments in a cluster-randomized 2x2 trial with 1,560 maize-growing households in eastern Uganda: seed trial packs, which let farmers experience production traits on their plots, and cooking demonstrations with blind tasting, which expose them to consumption traits. Both raise farmers' knowledge and use of the promoted hybrid. Trial packs generate the larger increase, but it runs almost entirely through farmers recycling grain saved from the trial rather than buying fresh seed, a privately rational choice given high seed prices, uncertain quality, and liquidity constraints. Because the variety is a hybrid, recycled seed loses vigor over successive seasons; whether the pack sustains turnover then hinges on farmers returning to the market once their stock degrades, plausible given the durable gains in awareness and intent we document, but unobserved here. Cooking demonstrations produce a smaller increase that runs through fresh-seed purchases, with suggestive yield gains. Trial packs are cheaper per adopter, and the lesson is to match instrument to technology: packs deliver turnover directly for recyclable open-pollinated varieties, while for hybrids their long-run payoff runs through repeat re-purchase and so depends on an accessible seed market.

Keywords: technology adoption, consumption and production traits, trial packs, cooking demonstration and blind tasting, maize, Uganda.

JEL: Q16, O33, Q12, D83, C93

1 Introduction

To sustainably feed a growing population while addressing climate change and preserving biodiversity, it is essential to produce more food using less land (Tilman et al., 2011). Green Revolution technologies, particularly improved crop technologies, have been key to achieving higher yields and enhancing resilience to environmental challenges,

[*]Innovation Policy and Scaling Unit, International Food Policy Research Institute, Leuven, Belgium
[†]One Acre Fund, Jijna, Uganda
[‡]Markets Trade and Institutions Division, International Food Policy Research Institute, Nairobi, Kenya
[§]Markets Trade and Institutions Division, International Food Policy Research Institute, Washington DC, USA

such as drought, and emerging biotic stresses, such as pests and diseases (Evenson and Gollin, 2003). Realizing these gains in farmers' fields, however, requires varietal turnover: the replacement of older varieties with newer ones that embody recent genetic improvements. In much of sub-Saharan Africa, turnover is slow, and many farmers continue to grow varieties released decades ago, so that recent genetic gains are not reflected in the seed actually planted (Colen et al., 2024). How to accelerate varietal turnover is therefore a central question for seed-system policy. Traditionally, crop improvement efforts in breeding programs which serve smallholder farmers have focused on improving production-related agronomic traits, such as yield, drought tolerance, disease resistance, and pest resistance. However, as farmers in sub-Saharan Africa often produce for self-consumption (Carletto, Corral, and Guelfi, 2017; Bellon et al., 2020), the adoption of new crop varieties is also influenced by consumption-related traits, such as taste, color, texture, and ease of cooking (Timu et al., 2014; Thiele et al., 2021; Otieno et al., 2011). Integrating improvements in such traits alongside agronomic traits may be crucial for driving demand for improved crop varieties.

Because the production and consumption attributes of new varieties are not directly observable to farmers upon seed purchase, widespread adoption also may require raising farmers' awareness of their beneficial attributes. A common strategy to encourage farmers to learn about a new variety is through the distribution of seed sample packs, often referred to as "trial packs", "starter packs", or "mini-kits" (Blackie and Mann, 2005; Cromwell, 1990). These seed trial packs typically contain small quantities of seed (e.g., 1 kg) and are provided for free, allowing farmers to test them on a small portion of their plot.[1] The primary aim is to reduce the cost and risk associated with learning about the production attributes of the variety, such that farmers will purchase the seed in subsequent seasons if they find doing so beneficial.

Strategies that explicitly seek to raise farmers' awareness of the consumption-related attributes of improved varieties are less common. This is perhaps surprising given the theoretical and empirical literature showing that in smallholder agriculture settings consumption and production decisions are intricately related (Singh, Squire, and Strauss, 1986; LaFave and Thomas, 2016). Moreover, interventions that familiarize farmers with consumption traits, such as cooking demonstrations and tasting events, are often primarily concerned with nutrition-related outcomes and rarely consider actual adoption of the crop or variety that was used during the demonstration (Reicks, Kocher, and Reeder, 2018).[2]

This paper asks which of two contrasting extension instruments most cost-effectively accelerates varietal turnover among smallholder farmers, recognizing their dual role as both producers and end-users: one targeting the production side of variety choice, the other the consumption side. We test their relative effectiveness, and their cost per additional adopter, in a cluster-randomized control trial. The first intervention targets production traits by providing farmers with 1 kg trial packs of seed of an improved variety.[3] A second intervention involves cooking demonstrations and blind tasting tests. This intervention allows farmers to familiarize themselves with consumption attributes of maize flour derived from the grain grown using the same improved variety that was promoted in the seed trial pack intervention. Additionally, farmers in this treatment arm are provided with maize flour samples derived from the grain grown using the promoted improved variety to further explore consumption traits at home and potentially convince other household members of consumption related benefits.

The field experiment was conducted in four districts in eastern Uganda among a representative sample of about

[1]Some trials may also include small quantities of other relevant inputs such as fertilizer, pesticide, and/or herbicide. For instance, Malawi's starter pack program provided both seeds and fertilizer (Blackie and Mann, 2005).

[2]While seed trial packs may also enable farmers to discover consumption traits after they harvest the grain obtained from the seed trial pack, such learning is often not the primary focus of trial packs and may occur only if farmers taste the harvest from the trial pack before mixing it with other varieties and/or selling it.

[3]We use the term "improved variety" throughout this paper to refer to both fresh maize hybrids and open-pollinated varieties (OPVs) that are either newly purchased or have been appropriately recycled (up to four seasons). This excludes farmer-saved seed or seed from informal exchanges, which, in the specific context of maize, may be less effective due to cross pollination and genetic drift over multiple generations, or due to poor seed storage and handling between seasons (Pixley and Banziger, 2004; Roos, 1980).

1,500 farmers. The area, commonly known as the Busoga Kingdom, is predominantly populated by smallholder farmers who cultivate maize for both home consumption and sale. The study ran over two consecutive agricultural seasons in 2023, with endline data collected in early 2024 after the harvest of the second season. The variety we promoted in this study is a hybrid marketed as *Bazooka* (UH 5354), which is widely available from agro-dealers but has yet to see broad adoption among farmers. *Bazooka*'s primary notable production trait is its yield potential, which is over double that of popular open-pollinated varieties (OPVs) such as Longe 5 (Semalulu et al., 2022). However, it also offers important consumption attributes: *Bazooka* combines high starch content with mild sweetness and produces a very white porridge.

We find that farmers who received a trial pack are substantially more likely to plant *Bazooka* in the subsequent season. This increase, however, is driven almost entirely by the reuse of grain harvested from the trial pack rather than by purchases of fresh *Bazooka* seed. When we restrict attention to the use of fresh *Bazooka* seed, adoption rates are statistically indistinguishable between treatment and control farmers. Moreover, when we apply a broader but agronomically valid definition of seed of improved varieties that includes fresh hybrids and fresh or appropriately recycled OPVs (but exclude recycled hybrid grain), we find that the trial pack intervention reduces the use of seed sourced from the commercial market. This reflects substitution rather than a net loss of improved-variety use: treated farmers, having access to *Bazooka* grain at no monetary cost, become less likely to purchase fresh hybrid seed or fresh or appropriately recycled OPV seed in the following season.

For the cooking demonstration and tasting intervention, we find suggestive evidence consistent with an increase in adoption of fresh *Bazooka* seed.[4] We find no evidence that either intervention influenced how the harvested maize in the next season was used—whether for own consumption, market sales, or saved to be used in the next season.

We see little evidence that receiving both treatments had additional impacts beyond those of the main treatment—this may have been the case if, for instance, some household members are more sensitive to production traits and others are more sensitive to consumption traits, and the trial pack facilitated coordination—though the estimates are fairly noisy. Contrary to our expectations, highlighting consumption traits did not increase women's involvement in seed-related decision-making. Finally, we observe no downstream impacts of either intervention on household food security or overall welfare.

Exploring some of the impact pathways, we do find that the cooking demonstration and blind tasting increased the share of farmers who rank improved varieties higher on a range of consumption attributes such as taste, portion size, appearance, and ease of cooking. Furthermore, we find that farmers who received the seed trial pack rank improved varieties higher in terms of production-related attributes such as yield, abiotic and biotic stress resistance, time to maturity, and germination rate, although the evidence is less convincing. Interestingly, the seed trial pack also positively affects how farmers think about consumption traits of improved varieties, suggesting farmers also consume at least part of the grain obtained from the trial pack and pay close attention to consumption attributes.

A key finding from our study is that the majority of farmers who received the seed trial pack continued using the distributed hybrid maize variety by recycling the harvest from the trial pack rather than purchasing fresh seed. This behavior does not seem to be merely the result of a lack of knowledge about the drawbacks of recycling hybrid seeds; rather, it may represent a rational response to the conditions farmers face: Even when farmers are aware of potential yield penalties associated with reusing hybrid seed, they may still find recycling preferable in light of high cost of seed, liquidity constraints, market risk, and distrust in agro-dealer networks. These findings suggest that exposure alone is not enough to ensure continued use of fresh improved seed, and that addressing the informational, economic, and structural barriers that shape farmers' decision environments is essential for sustained adoption.

Setting these effects against the actual cost of delivering each instrument, we find that the most cost-effective instrument depends on the policy objective. If the goal is varietal turnover, seed trial packs are far cheaper per additional adopter than cooking demonstrations. If the goal is to build demand in the commercial seed market,

[4] We deem this evidence as suggestive because, while false discovery rate–adjusted q-values point to significant effects, our prespecified analysis, which aggregates outcomes into indices across related measures, does not show significant effects.

the ranking reverses: trial packs generate turnover through recycling and leave fresh-seed purchases unchanged, whereas cooking demonstrations are the only instrument that moves this margin, alongside suggestive yield gains—though within our one-season window such purchases reflect market entry rather than established repeat demand. Because trial packs are almost entirely variable cost while demonstrations carry a large per-session cost shared across attendees, this ranking is sensitive to how many farmers each demonstration reaches. More fundamentally, because the promoted variety is a hybrid that loses vigor when recycled, whether the pack sustains turnover depends on farmers re-buying once their recycled stock degrades; the durable gains in awareness and intent it produces make this plausible, but our one-season design cannot observe it.

Our work contributes to several strands of the literature on agricultural technology adoption. First, we add to the literature examining whether short-term input subsidies—whether provided in cash or in-kind—can stimulate longer-term uptake of improved technologies (Macours, Mallia, and Rudder, 2025; Balew, Bulte, and Kassie, 2025; Gignoux et al., 2023; Fishman et al., 2022; Carter, Laajaj, and Yang, 2021). Specifically, despite the prevalent use of temporary subsidies as a policy instrument to spur agricultural technology adoption (Christinck, Diarra, and Horneber, 2014; Dorward and Kydd, 2005; Phiri et al., 2000), we add to a relatively limited set of experimental impact evaluations testing the effectiveness of trial packs in accelerating technology adoption, which finds mixed results.[5] For instance, Emerick et al. (2016) distribute starter kits for seed of a new rice variety in India and find that seventy-six percent of treatment farmers cultivated the technology in the second season following distribution. On the other hand, Biedny et al. (2020) find that, in Tanzania, adding trial packs to demonstration plots run by village-based agricultural advisors had no significant impacts on sales, orders received, or learning outcomes, and Maredia et al. (2025) further show no increases in adoption at the 1 or 4 year marks as a result of trial pack provision. Similarly, Jones et al. (2022) find provision of a horticulture mini-kit to farmers in Rwanda has no significant impact on the adoption of horticulture. Ragasa, Oyinbo, and Ma (2025) present results from Nigeria showing that seed trial packs substantially increase adoption of improved maize and cowpea OPVs.

We also contribute to the literature on consumer acceptance of improved varieties, by introducing an intervention that exposes farmers to a variety's consumption traits. While many studies use consumer tasting of new varieties as part of their research, in order to either assess consumer preferences for the varieties or sensitize consumers to varietal traits before eliciting willingness to pay (Birol et al., 2015; De Groote et al., 2014; Okello et al., 2021), these studies do not treat the tasting experience as an intervention designed to induce take-up of new varieties. As such, the opportunity to try new variety is not randomized between participants. Existing interventions that expose farmers to consumption attributes of new varieties also tend to promote nutritional (credence) traits, rather than experience traits such as taste or cooking quality, and rarely measure subsequent adoption in the following season (Olney et al., 2015; Kramer, 2017; Schreinemachers, Patalagsa, and Uddin, 2016; Osei et al., 2017; Murty, Rao, and Bamji, 2016). One exception is De Brauw et al. (2018), who study the impact of the HarvestPlus' Reaching End Users project (which includes a demand creation component that focuses on consumption attributes) on the adoption of bio-fortified sweet potato in Mozambique and Uganda. They find that the combination of the demand creation treatment and production-focused extension activities increases adoption of vitamin A fortified orange-fleshed sweet potatoes in the subsequent season by over 60 percent in both Mozambique and Uganda.

Finally, we contribute by comparing production- and consumption-oriented extension instruments not only on their effectiveness but on their cost. To our knowledge, this is the first study to price two such instruments against each other using their actual delivery budgets, and to show that the most cost-effective choice depends on whether the objective is varietal turnover or the development of a commercial seed market. This shifts the question from

[5]While other studies with interventions featuring a trial pack exist, the aim of many is not to evaluate its impacts on future technology adoption. For instance, Boucher et al. (2024) provide a trial pack of drought tolerant varieties to farmers before offering households the opportunity to purchase the variety bundled with index insurance in the subsequent season. Other studies similarly look at the effect of trial pack provision on other outcomes, such as willingness-to-pay for seeds (Morgan, Mason, and Maredia, 2020) or beliefs about the yields of improved varieties (Tjernstrom and Gars, 2019) .

whether a given instrument works to which instrument a budget-constrained program should choose, and at what scale.

The remainder of the paper is organized as follows. Section 2 outlines the research methodology, including the experimental design and empirical strategy. Section 3 describes the study context and data sources. Section 4 presents the main findings on use of the promoted variety, the source of the seed used, and production outcomes, along with secondary outcomes related to use of harvest output, intra-household decision-making, food security, and overall well-being. Section 5 investigates the mechanisms through which each instrument shifts adoption. Section 6 examines why farmers who receive a trial pack continue through recycling rather than purchasing fresh seed. Section 7 compares the two instruments on cost-effectiveness and scaling. Section 8 concludes.

2 Methods

2.1 Experimental Design

We use a cluster randomized control trial (RCT) to evaluate the effectiveness of two interventions, using a 2x2 factorial design, with treatments assigned at the village level to mitigate potential concerns about spillover effects. The first factor in the factorial design is the seed trial pack treatment: sampled farmers in the treatment villages received a free sample of a hybrid maize variety (*Bazooka*), whereas those in the control villages did not. The second factor involved a cooking demonstration and tasting session; in treatment villages, sampled farmers were invited to a session where they observed the preparation of maize flour made from the *Bazooka* hybrid variety, participated in a tasting session, and received a free sample of *Bazooka*-derived maize flour to try at home. No such activities were conducted in control villages.[6]

2.2 Treatments

The first intervention involves providing a seed trial pack to the household member responsible for most maize cultivation decisions, who in the study context is typically the male co-head (Van Campenhout, Lecoutere, and Spielman, 2023). This trial pack contains *Bazooka* maize, which was released by the National Agricultural Research Organization (NARO) in Uganda in 2013 and is currently available in the market but not yet widely adopted by farmers.[7] Some of *Bazooka*'s key features include high yield, resistance to drought, and resistance to diseases like maize lethal necrosis (Akwango-Aliau et al., 2022).[8] Specifically, we provided 1 kg bags of *Bazooka* seed, which is enough to plant about one-eighth of an acre. Given that the average farmer in our sample cultivates 1.24 acres of maize, this seed covers approximately 10 percent of their total cultivated area.

The second intervention consists of a cooking demonstration and tasting event, organized once in each village assigned to the cooking demonstration treatment. Within these villages, all sampled households were invited to attend a facilitated meeting at a central location (half of whom were cross-randomized to receive a seed trial pack under the factorial design). As for the cooking demonstration and tasting intervention, invitations were addressed to the household member responsible for maize production decisions. Invitees were explicitly encouraged to attend together with another household member involved in food preparation, in order to ensure that both production and consumption perspectives were represented during the session.

During the event, a facilitator started by asking the group to mention the most commonly grown maize varieties

[6] Farmers in pure control villages under both factors received a "token of appreciation" of similar value to the seed trial pack/maize flour bag to account for potential income effects.

[7] At baseline, only about 8 percent of sampled farmers reported growing *Bazooka*—See Table 1 in Section 3.3 below.

[8] The primary production trait of *Bazooka* is its high yield, underscored by the package slogan, "yield explosion."

by farmers in their village. These varieties are grouped into “improved varieties” and “local varieties” on a flip chart.[9] Farmers were then asked to rank the two categories based on ratings of various consumption attributes by a show of hands. To facilitate the discussion, flip charts were pre-filled with the five consumption traits most commonly mentioned during focus group discussions held during the design phase of the study: taste, texture, color, aroma, and the degree to which the flour expands during cooking. Farmers could add as many additional traits as they saw fit.

After the rating exercise, sessions proceeded with the cooking demonstration and blind taste testing. The facilitator asked a volunteer participant to prepare *posho*, a thick, dense staple porridge made by mixing maize flour with boiling water until it reaches a dough-like consistency. The volunteer cook then prepared two meals: one using flour obtained from a local variety and one using flour derived from *Bazooka*. Neither the cook nor the other participants were aware of which flour corresponded to which maize type. The research team provided all necessary utensils for the session, including a gas stove, pots, aprons, and even a chef’s hat. We ensured that the two dishes differed only in terms of the flour used by employing the same cook and starting with identical amounts of flour, measured on a weighing scale. The resulting dishes were displayed on a table, and participants were invited to taste the two varieties, simply labeled by their position as the variety on the “left” and the variety on the “right.” The participants then rated the two varieties on the various consumption attributes again, voting on which of the two samples is superior for each attribute by a show of hands.

The rating results were then discussed within the group and participants were informed that one sample was made from flour obtained from local maize while the other was made from an improved maize variety called *Bazooka*. After being asked to guess which was which, the facilitator gathered the guesses and revealed the correct answers. As the results in Section 5.2.1 will show, improved seed varieties are generally perceived favorably with respect to consumption traits, yet these positive perceptions are far from universal, leaving scope for further improvement. At the end of the session, participating households received a 2 kg take-home sample of maize flour produced from the *Bazooka* variety, sufficient for preparing at least one family meal, to facilitate post-session experimentation at home.

In both treatments, facilitators followed a standardized script and were instructed to avoid promotional language, focusing instead on eliciting participants’ own assessments and facilitating discussion rather than advocating for a specific variety. The objective of both interventions was to study learning about improved crop technologies in general, rather than to promote adoption of a particular seed. Accordingly, discussions in both treatments were kept deliberately broad, emphasizing generic attributes of improved maize varieties, such as yield potential and consumption traits, rather than the merits of *Bazooka* per se.

At the same time, the underlying technology used to illustrate these attributes was held constant across interventions. In the seed trial pack treatment, the distributed seed was clearly labeled as *Bazooka*, while in the cooking demonstration, the maize flour used for preparation was explicitly identified as being derived from *Bazooka* seed. By maintaining transparency about the source technology while keeping the framing comparable across treatments, we aimed to ensure that differences in outcomes across interventions cannot be attributed to differences in the underlying technology itself, but rather to the mode of learning emphasized by each intervention.

The two interventions deliberately differ in how exposure within the household was structured. The seed trial pack intervention targets the household member responsible for maize production decisions, while the cooking demonstration and tasting intervention was designed to involve both production- and consumption-relevant decision-makers by encouraging joint attendance. This design aligns each intervention with the margin it is intended to affect: production-oriented learning through agronomic experimentation on the one hand, and consumption-oriented learning through sensory experience on the other. By structuring exposure in this way, the design leverages existing household decision-making roles rather than confounding treatment effects with arbitrary differences in which

[9] The terms for the seed types in the local language were “Dhuuma Omulongosemu” for “seed of an improved variety” and “Dhuuma Omusoga” for local seed. The latter is derived from the name of the region, Busoga.

household members were exposed.

2.3 Estimation and Inference

We estimate intent-to-treat (ITT) effects using linear regression models that compare mean outcomes across treatment and control groups. Given that randomization was conducted at the village level, we will estimate ITTs using the following equation:

$$Y_{ij} = \alpha + \beta_S T_j^S + \beta_D T_j^D + \beta_I T_j^S T_j^D + \delta Y_{ij}^B + \varepsilon_{ij} \tag{1}$$

where T_j^S is an indicator variable that equals one if village j was randomly assigned to the seed trial pack intervention (and zero otherwise), and T_j^D is an indicator variable that equals one if village j was randomly assigned to the cooking demonstration and blind tasting intervention (and zero otherwise). Outcomes are measured at the individual level (Y_{ij}). We allow for an interaction effect between the two interventions, and we control for baseline outcomes $\left(Y_{ij}^B\right)$ to improve precision. We apply a cluster-robust variance estimator with the bias-reduced linearization (CR2) small-sample correction (Imbens and Kolesar, 2016), with standard errors clustered at the level of randomization (village level).

The coefficients in equation 1 map directly onto our hypotheses. The central hypothesis is that each instrument raises subsequent use of the promoted variety, so we expect $\beta_S > 0$ for the seed trial pack and $\beta_D > 0$ for the cooking demonstration. Because the two instruments target different margins of the same decision, the trial pack the production side and the demonstration the consumption side, we have no strong prior on which effect is larger, and comparing β_S with β_D is itself one aim of the study. The interaction β_I tests whether receiving both instruments adds to the sum of their separate effects: it would be positive if production- and consumption-side learning are complements, for example if different household members respond to each, and close to zero if they are redundant. Finally, we expect the trial pack to raise use of the promoted variety mainly through recycling rather than fresh purchases, so that β_S is positive for use of the variety overall but close to zero for use of fresh seed, whereas the demonstration, which distributes no seed, can raise use only through fresh purchases.

Factorial designs are commonly employed to evaluate multiple treatments within a single experiment. While our main specification is the fully interacted model in equation 1, our pre-analysis plan also states that we may wish to enhance statistical power by pooling observations across the orthogonal treatments if we find that a treatment effect is smaller than the minimal detectable effect size assumed during power calculations (Colen et al., 2024). For conciseness, we include results from these pooled specifications, which are qualitatively similar to the results from the fully interacted model, in Online Appendix A1 rather than the main text.

While our main outcome of interest is whether individuals adopt improved varieties the subsequent season, we also look at additional outcomes related to productivity, use of harvest output, intrahousehold decision-making, and household welfare. Since we evaluate treatment effects across a range of outcomes, it is necessary to address the multiple comparisons problem. As prespecified, we address the issue following Anderson (2008) and aggregate various outcome measures within a given family into summary indices. Each index is calculated as a weighted average of the standardized (z-score) values of the individual outcomes. The weights are derived from the inverse of the covariance matrix of the outcomes within the family, such that outcomes that are more highly correlated with others receive less weight. This efficient generalized least squares (EGLS) estimator maximizes the statistical power of the index by down-weighting redundant information and emphasizing variation that is orthogonal across outcomes. However, because it may also be useful to understand changes in specific outcomes within a given family rather than just changes in index values, we also present the coefficients on each outcome, controlling the false discovery rate (FDR) by using sharpened two-stage q-values proposed by Benjamini, Krieger, and Yekutieli (2006). In each results table, the summary index reported in the top row aggregates the individual outcomes listed in the rows beneath it, so the composition of each family can be read directly from the corresponding table. Because

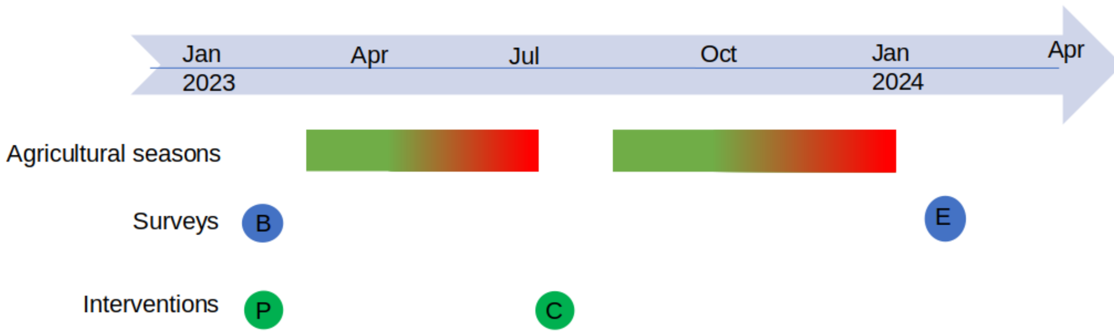

Figure 1: Timeline

outcomes are demeaned and scaled using control-group moments before aggregation, the indices have mean zero in the control group by construction (up to rounding).

2.4 Timeline

The experiment spans two consecutive agricultural seasons in order to accommodate the inherently different durations of the two interventions while measuring outcomes at a common point in time. For the seed trial pack treatment, the outcome of interest is not (the first stage result) whether farmers plant the trial pack itself, but whether the experimentation enabled by planting the trial pack in one season leads to continued use of the variety in the subsequent season. This necessarily requires observing outcomes after an additional planting cycle.

In contrast, the cooking demonstration and tasting session is a one-time informational intervention that does not require a full growing season to implement. To ensure comparability, this intervention was therefore administered at the beginning of the second experimental season, such that outcomes for both interventions are measured after the same agricultural season.

The study area experiences two maize growing seasons per calendar year. The first season, locally known as *Entoigo*, runs from March/April to June/July, while the second season, *Nsambya*, extends from August/September to November/December. *Bazooka* can be cultivated in either season. As illustrated in Figure 1, seed trial packs were distributed alongside baseline data collection prior to the start of *Entoigo* 2023, allowing farmers to plant the trial pack during that season. Cooking demonstrations and tasting sessions were conducted after the completion of *Entoigo* and before the onset of *Nsambya*. Endline data were collected in February 2024, following the conclusion of *Nsambya*.

Importantly, while the two interventions differ in their duration of exposure—season-long agronomic experimentation versus a short informational session—all outcomes are evaluated at the same point in the agricultural cycle, ensuring that estimated treatment effects are not driven by differences in outcome measurement timing.

3 Study Sample and Data

3.1 Sample Size Calculation

Sample size was determined through a series of power simulations detailed in the pre-registered pre-analysis plan (Colen et al., 2024). The primary outcome used in these simulations is a binary indicator for the type of seed used by the farmer (1 if the farmer used the prompted seed variety, 0 otherwise). Data from a previous project, which included 3,450 smallholder maize farmers across 345 villages, were used to estimate the intra-cluster (within-village) correlation for this outcome (Miehe et al., 2023). We assumed a 13.5–percentage point increase for both the seed sample treatment and the cooking demonstration treatment, and a 23.5–percentage point increase for the interaction effect as minimal detectable effect size.

The simulations resulted in a sample design of 148 villages, with 10 households sampled per village, assigned evenly across the four cells of the 2×2 factorial design. Specifically, 37 villages were assigned to each of the following groups: pure control, seed trial pack only, cooking demonstration only, and both interventions. This implies that, in total, 74 villages received the seed trial pack (those assigned to the seed-only and both-treatment groups), and 74 villages were exposed to the cooking demonstration and tasting sessions (those assigned to the demo-only and both-treatment groups). The resulting sample therefore comprises 1,480 households across 148 villages. To counteract potential attrition which would decrease power, we increased the sample size by adding 2 additional villages in each treatment cell. This leads to a total sample size of 1,560 households located in 156 villages.

3.2 Sampling Strategy

The study was conducted in Eastern Uganda in an area that is also known as the Busoga Kingdom. The main income-generating activity in this area is smallholder maize production. We selected our sample from four districts (Bugiri, Iganga, Kamuli, and Mayuge) chosen for their relatively low adoption rates of *Bazooka,* despite a well-established network of agro-input dealers.

To obtain a representative random sample of farmers in the Busoga Kingdom, villages were selected with probability proportional to the number of households within each village, and then randomly assigned to one of the four study arms (control, trial pack only, consumption demonstration only, both trial pack and consumption demonstration). Within each sampled village, 10 households were randomly chosen to participate in the study from the list of farm households maintained by the village council.

3.3 Descriptive Statistics and Baseline Balance

We pre-registered 10 variables to assess balance in our design during baseline data collection. These variables were selected to offer a comprehensive description of a representative farmer in our sample. Five of the variables are characteristics that are unlikely to be influenced by the intervention, while the other five are drawn from the pre-registered primary and secondary endline outcomes.

Table 1 shows control means in the first column (and standard deviations below). We see that the average household head in the control group was about 50 years old at the time of the baseline survey. About half of the household heads had finished primary education, and in 16 percent of the households, the household head was a woman. Households in the area are large, with on average 8 to 9 members. The average distance to the nearest agro-input shop where maize seed of an improved variety can be bought is about 4 kilometers.

In terms of variables that will be used to assess impact at endline, we first inquired whether farmers had used *Bazooka*, the hybrid seed variety that is also utilized in our experiment. At baseline, only about 8 percent of farmers in the control group reported having used *Bazooka* seed on at least one plot in the previous season (*Nsambya* of 2022). We also asked a less specific question where we asked if "quality maize seed, such as an OPV or hybrid seed" was used on any of their plots during the previous season. Approximately 42 percent of control group households answered affirmatively to this question at baseline.

We also asked where the seed used on a randomly selected plot was obtained from.[10] Results indicate that

[10]During both baseline and endline, detailed plot-level questions (such as seed source, price, etc.) were collected for a single, randomly selected plot per household to limit respondent burden. If trial pack seed is systematically planted on particular plots (for example,

approximately 29 percent of control farmers sourced their seed from formal channels, such as agro-input dealers, non-governmental organizations, or the government extension system. Conversely, 52 percent of control group farmers reported reusing seed from previous seasons, with some having used it for more than four seasons despite recommendations against using hybrid varieties more than once and against using OPVs more than four times, at the risk of losing any yield advantage. Finally, the average farmer in the control group harvested about 380 kilograms per acre on the randomly selected plot at baseline.

The table reports baseline mean differences between the trial pack group and the control group obtained from the fully interacted specification in Equation 1 (column 2, β_S), between the cooking demo group and the control group (column 3, β_D), and the interaction term capturing the additional effect of receiving both treatments beyond the sum of individual effects (column 4, β_I). Overall, the results suggest good baseline balance, as none of the treatment–control differences are statistically significant.

3.4 Attrition

Attrition was minimal: during endline, we successfully located all but four of the 1,560 households interviewed at baseline. Moreover, only 18 of the remaining 1,556 households could not be interviewed due to circumstances such as burial or illness, or they refused participation, reducing the effective sample size at endline to 1,538, or 98.6% of the baseline sample. Attrition was balanced across treatment arms.

3.5 Measurement

All endline outcomes are self-reported, which raises two measurement concerns. First, farmers may misreport which variety they grow: validation studies comparing self-reported varietal identity against DNA fingerprinting document substantial misclassification in smallholder settings (Kosmowski et al., 2019). This concern is limited for the trial-pack arm, where the variety planted is the one we distributed and recall of receipt is near-universal, but it remains for the control group and for the broader improved-seed measures. Because non-differential misclassification of the outcome generally attenuates estimated effects toward zero, we regard our adoption estimates as conservative. Second, self-reports may be subject to experimenter demand effects, with treated farmers over-reporting use of the promoted variety. Two features limit this concern: our first-stage measures are corroborated by administrative distribution and attendance records rather than recall alone, and the overall pattern of results is hard to reconcile with simple demand bias, since treated farmers report recycling rather than the fresh-seed purchases the intervention encouraged, and we find no effects on outcomes such as food security or women's involvement where such bias would also be expected. We nonetheless interpret the self-reported magnitudes with appropriate caution.

4 Results

We now turn to the main results, with subsections for first stage results, adoption of improved crop technologies, use of harvest, women's empowerment, and welfare and food security.

Table 1: Baseline balance

	control mean	ITT trial pack	ITT cooking demo	ITT interaction
Age of household head (in years)	49.71	-1.53	-1.87	2.45
	(13.29)	(1.31)	(1.22)	(1.69)
Household head has finished primary education (1=yes)	0.51	0.00	-0.02	0.04
	(0.50)	(0.05)	(0.04)	(0.06)
Gender of household head (1=male)	0.84	-0.03	-0.04	0.00
	(0.37)	(0.03)	(0.03)	(0.05)
Household size	8.36	-0.05	-0.03	0.05
	(3.71)	(0.04)	(0.04)	(0.06)
Distance of homestead to nearest agro-input shop (km)	4.10	0.01	0.06	-0.05
	(3.84)	(0.14)	(0.15)	(0.19)
Has used the promoted seed (*Bazooka*) on any plot in last season (1=yes)	0.08	0.04	0.03	-0.07^{+}
	(0.27)	(0.03)	(0.03)	(0.04)
Has used quality maize seed on any plot in last season (1=yes)	0.42	-0.03	-0.01	-0.02
	(0.49)	(0.05)	(0.04)	(0.07)
Seed on random plot was obtained from formal seed source (1=yes)	0.29	-0.01	0.05	-0.06
	(0.45)	(0.04)	(0.04)	(0.06)
Used seed that is d more than 3 seasons on randomly selected plot (1=yes)	0.52	0.00	0.02	0.04
	(0.50)	(0.05)	(0.05)	(0.07)
Maize yields on a randomly chosen plot in last season (kg/acres)	384.37	-0.04	0.02	-0.04
	(286.19)	(0.07)	(0.07)	(0.11)

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels. Household size, distance to nearest agro-input shop, and productivity is transformed using the inverse hyperbolic sine transformation (means and standard deviations are in levels).

Table 2: First Stage Results

	Crtl	Seed trial pack
At endline, household recalls being given seed trial pack (%)	8.6	98
Household received seed trial pack (%)	–	100
Trial pack planted in *Entoigo* 2023 (%)	–	99
Median area planted with trial pack (acres)	–	0.25
	Crtl	Cooking demo
At endline, household recalls being invited for cooking demo (%)	4	92
Household attended cooking demo and tasting session (%)	–	89
Both male and female co-heads attended (%)	–	48
Used maize flour (1=yes) (%)	–	99

4.1 Compliance and Treatment Recall

Self-reported recall of the treatments was high for both interventions (Table 2). During the endline survey, we asked all farmers whether they recalled receiving the interventions, regardless of assignment. This provides useful information on failure to reach intended beneficiaries and on spillovers or exposure among non-targeted households. For the trial pack treatment, 98 percent of farmers in the treatment group indicated that they received a seed trial pack from us in March 2023. Furthermore, 91.4 percent of farmers in the control group indicated that they had not received a seed pack from us. For the cooking demonstration treatment, 92 percent of farmers in the treatment group recalled being invited to a cooking and tasting demonstration. Meanwhile, 96 percent of farmers in the control group did not recall a cooking and tasting demonstration invitation.

Administrative records from the seed trial pack distribution, which happened concurrently with baseline data collection, show that no households that were assigned to this treatment refused the seed. Consistent with this, nearly all households who recalled receiving the trial pack reported planting it during the first experimental season (*Entoigo* 2023). Among households that planted the trial pack, the median area planted was 0.25 acres, approximately twice the nominal one-eighth acre coverage implied by the size of the trial pack.

Households that were assigned to the cooking demonstration and tasting sessions received an invitation; attendance was voluntary. Administrative attendance records indicate that just under 90 percent of households assigned to the cooking demonstration intervention also attended. Among attendees, approximately half participated jointly as male and female co-heads. Endline survey responses further indicate that nearly all households who reported attending the demonstration also reported using the maize flour provided at the end of the session to prepare a meal at home, rather than selling or otherwise disposing of it. Together, these figures indicate high intervention fidelity and substantial treatment uptake, supporting an intent-to-treat interpretation of the estimated effects.

4.2 Adoption

Adoption of the promoted seed in the season following the intervention is the primary outcome of interest in this study. Accordingly, Table 3 examines whether farmers used the promoted variety at all, while Table A11 reports more disaggregated outcomes on seed use and management practices measured on a single randomly selected maize plot within each household. As described in Section 2.3, related outcomes are aggregated into summary indices following Anderson (2008).

smaller plots), random plot selection leads to misclassification of plot-level treatment status for some treated households. This attenuates estimated effects toward zero relative to the effect on the plot that actually received the trial pack. We therefore interpret plot-level results as conservative lower bounds on effects at the plot that actually receives the trial seed. However, because both treatment assignment and plot selection are randomized, the estimate remains unbiased for the intent-to-treat effect on a randomly selected plot.

The structure of both tables mirrors that of Table 1. Column (1) reports control group means at endline, while columns (2) to (4) present intent-to-treat estimates from the fully interacted specification in Equation 1, corresponding to the trial pack intervention, the cooking demonstration intervention, and their interaction, respectively.

Turning to the summary adoption index in Table 3, we find a positive and statistically significant intent-to-treat effect of the trial pack intervention. In contrast, the tasting session intervention has no detectable effect on the index. We also find no evidence of an interaction effect between the two treatments, suggesting that their combined provision does not generate additional adoption beyond the effect of the trial pack alone.

We next consider the individual outcome components underlying the adoption index. We first look at whether households planted the promoted variety, *Bazooka*, on at least one maize plot. In the control group, approximately 10 percent of farmers report having grown *Bazooka* in the season that followed the season where they could experiment with the seed from the trial pack. Receipt of a trial pack increases this likelihood by 67 percentage points, corresponding to an adoption rate of roughly four out of five farmers in the treatment group. The point estimate for the cooking demonstration and tasting session is also positive (p=0.051), but it is not statistically significant after adjusting for multiple hypothesis testing using false discovery rate corrections (q=0.110). We likewise find no evidence of interaction effects between the two interventions.

Moving on to measures of intensity of adoption of the promoted variety, we find that the trial pack intervention increases not only the likelihood of planting *Bazooka* but also the scale of its use. Households receiving a trial pack plant *Bazooka* on more plots, allocate a larger share of maize plots to the variety, and devote more land area to its cultivation, both in absolute terms and as a share of total maize area. All of these effects are positive and statistically significant. For the orthogonal treatment, coefficient estimates are also positive, but they again do not survive adjustment for multiple outcomes. Interaction effects are absent.

Because recycled hybrid grain is not agronomically recommended, as it does not retain the genetic characteristics of the original hybrid and typically entails substantial yield penalties in subsequent generations, we also consider a narrower adoption measure that defines adoption as the use of fresh *Bazooka* seed sourced from a trusted supplier (such as an agro-input dealer, the government extension system, or non-governmental organizations). In the control group, only about 5 percent of farmers report using fresh *Bazooka* seed on at least one plot. In contrast to when we allowed for recycling, we now find no evidence that receipt of a seed trial pack increases this share. Interestingly, under this narrower and agronomically grounded definition of adoption of the promoted variety, exposure to the cooking demonstration and tasting session is associated with a statistically significant increase in the likelihood that farmers use fresh *Bazooka* seed in the subsequent season, an effect that was not detectable under a definition that allowed for recycling of the promoted variety.

While the promoted variety is *Bazooka*, the objective of the intervention and of this study is not limited to the adoption of a single seed variety, but more broadly to the uptake of improved crop technologies, including both hybrid seed and open-pollinated varieties (OPVs). Accordingly, we also consider a broader, agronomically valid definition of adoption of seed of improved varieties that includes the use of fresh hybrid seed and fresh or appropriately recycled OPVs (but excluding recycled hybrid grain). Under this broader definition, we find that the trial pack intervention reduces the use of seed sourced from the commercial market. This reflects substitution rather than a net loss of improved-variety use: treated farmers become less likely to use fresh hybrid seed or fresh or appropriately recycled OPV seed, most likely because freely available recycled Bazooka grain is a strong substitute for purchased fresh hybrids and fresh or recycled OPVs.

Results on a randomly selected plot, reported in Appendix Table A11, are very similar: the seed trial pack produces a large increase in both the use and the intensity of cultivation of the promoted variety, the cooking demonstration a smaller positive effect (significant only at the 10 percent level), and there are again no interaction effects.

The plot-level data add one result not visible at the household level. Despite the trial pack's large effect on use, we detect no impact on maize production or yield on the randomly selected plot, whereas the cooking demonstration

Table 3: Adoption

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Adoption Index	0.00	0.37**	0.09	-0.08	1495
	(0.81)	(0.07)	(0.08)	(0.10)	
Has used *Bazooka* on any plot in last season?† (1=yes)	0.10	0.67**	0.07	-0.08	1495
	(0.31)	(0.04)	(0.04)	(0.06)	
Number of plots planted with *Bazooka*	0.15	0.84**	0.07	-0.13	1495
	(0.48)	(0.06)	(0.05)	(0.09)	
Number of plots planted with *Bazooka* as share of total number of maize plots	0.10	0.65**	0.07	-0.11	1495
	(0.29)	(0.04)	(0.04)	(0.06)	
Area planted with *Bazooka* (acres)	0.13	0.75**	0.08	-0.15	1495
	(0.44)	(0.08)	(0.06)	(0.12)	
Area planted with *Bazooka* as a share of total maize cultivation area	0.10	0.65**	0.07	-0.11	1495
	(0.29)	(0.04)	(0.04)	(0.06)	
Has used fresh *Bazooka* on any plot in last season?† (1=yes)	0.05	0.02	0.05*	-0.01	1495
	(0.22)	(0.02)	(0.02)	(0.03)	
Has use fresh hybrid or OPV?† (1=yes)	0.30	-0.14**	0.03	0.02	1488
	(0.46)	(0.03)	(0.03)	(0.05)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). † indicates that baseline outcome was controlled for in the regression.

shows suggestive yield gains: imprecise in the fully interacted model, but statistically significant when pooled across the orthogonal treatment assignment (Appendix Table A11). This mirrors the source-of-seed split: households that recycle grain from the trial pack obtain no yield gain, whereas those who adopt through the purchase of fresh seed in the cooking demonstration arm show higher maize productivity. We return to why trial-pack farmers continue through recycling rather than purchasing fresh seed in Section 6.

The effect of the trial pack is concentrated among farmers with no prior experience of improved seed. Interacting the treatments with an indicator for baseline use of an improved variety (Appendix Table A13) shows that the trial pack raises use of the promoted variety by 72 percentage points among farmers who had not previously used improved seed, against 50 points among those who had, a difference significant at the 1 percent level, with a similar gradient for the adoption index. The trial pack thus expands adoption among the previously unreached rather than merely intensifying it among the already-initiated. We find no comparable heterogeneity for the cooking demonstration.

4.3 Use of Harvest

If the interventions affect how farmers perceive and use the promoted variety, they may also influence how harvested maize is allocated across competing uses. In particular, a focus on the use of commercial seed instead of replanting grain may affect how much of the harvest is retained for seed, while exposure to consumption attributes may increase the share allocated to own consumption or sale. We thus asked what farmers did with the maize that was harvested. To do so, we again focus on the randomly selected plot and ask how much was sold (or still planned to be sold), how much will be kept for seed and how much would be used for own consumption.[11] Table 4 summarizes treatment effects on these three outcomes (shares of maize consumed, sold, or held for seed) and an index thereof.

On average, households consumed the bulk of their maize harvest—about 75 percent—underscoring its central role in food security. Roughly 21 percent of maize was sold. Finally, a small portion of maize—just under 5 percent—was held for seed. Results show no meaningful shifts in post-harvest behavior attributable to the interventions.

4.4 Women's Empowerment

We also pre-registered the hypothesis that emphasizing consumption traits would raise women's influence over seed selection and the use of harvested maize, since in this setting men typically handle production and marketing decisions while women are more engaged in the consumption and cooking domain (Kramer and Trachtman, 2024; Van Campenhout, Lecoutere, and Spielman, 2023). We find no such effect. As Table A12 (Appendix) shows, women are already highly involved in these decisions at baseline, and neither intervention changes their involvement. Among demonstration attendees, adoption is no higher in households where both co-heads attended than where only one did, and if anything slightly lower; but this comparison is descriptive, since who attends is not randomly assigned.

4.5 Well-being and Food Security

Increasing the adoption of improved seed varieties is a means to an end; farmers ultimately aim to improve their households' food security and overall well-being. While it is unlikely that the small changes in yield that we observe for the cooking demonstration and tasting sessions would lead to meaningful changes in short-term welfare, we still evaluate the interventions' effects on key welfare indicators, including subjective well-being assessments, food security indicators, and a consumption expenditure aggregate. Results, reported in Online Appendix A2 show no impact.

[11] Note that the endline survey was implemented immediately after the harvest and so in many cases farmers did not sell yet.

Table 4: Use of harvest

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Use Index	-0.01	0.02	-0.01	0.08	1302
	(0.71)	(0.08)	(0.08)	(0.11)	
Share of maize consumed (%)	75.19	-1.21	-2.56	3.69	1449
	(29.97)	(3.00)	(2.80)	(4.21)	
Share of maize sold (%)	21.42	-1.50	-1.07	1.53	1495
	(29.98)	(2.93)	(2.67)	(4.16)	
Share maize kept for seed (%)	4.74	-0.24	0.11	-1.19	1321
	(7.93)	(0.86)	(0.86)	(1.21)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

5 Mechanisms

In this section, we examine intermediary outcomes that help connect the interventions to the adoption effects documented above. Specifically, we document changes in awareness of, and general attitudes toward, improved varieties, followed by changes in perceptions related to their consumption and production attributes.

5.1 Awareness About and General Attitudes Toward Improved Varieties

One reason farmers may have altered their planting decisions is that the interventions affected their awareness of, and general attitudes toward, the promoted seed. We explore this hypothesis using the family of outcomes presented in Table 5, which includes measures of varietal awareness, perception of riskiness, and general sentiment toward improved varieties. When summarizing these outcomes into an index in the top row of this table, we observe a significant positive impact of the seed trial pack on this set of outcomes. In contrast, the effect of the cooking demonstration and tasting session, while positive, is much smaller and statistically insignificant.

Looking at specific outcomes, we start by testing whether the interventions increased awareness of improved maize seed varieties, specifically *Bazooka*. We asked farmers to list as many improved maize seed varieties as they could name, and enumerators recorded the number of varieties mentioned. Second, we directly inquired whether the farmer was familiar with a maize variety called *Bazooka*. Table 5 suggest an impact on varietal knowledge resulting from both interventions. On average, control group farmers are aware of approximately 2 to 3 different maize varieties. Both interventions led farmers to name around 0.5 additional varieties. Overall, 65 percent of farmers in the control group report being familiar with the *Bazooka* variety. The seed trial pack intervention increases the probability that farmers report knowing about *Bazooka* by about 30 percentage points compared to non-trial pack group farmers. In contrast, the cooking and tasting demonstration shows a much smaller and insignificant 8 percentage point effect.

To capture more general attitudes toward improved varieties, we examine farmers' perceptions of downside risk associated with their use. Specifically, farmers were asked to assess how likely they believed it was that improved varieties would yield less than local varieties, a margin that is directly relevant for policies that distribute seed trial packs to lower perceived adoption risk. Responses were recorded on a four-point Likert scale ranging from "very likely" to "very unlikely." An analogous question was asked for the promoted variety, Bazooka, among farmers who reported familiarity with it. We construct an indicator for perceived downside risk that equals one if respondents selected either "very likely" or "somewhat likely." Overall, perceived downside risk is low, and we find no statistically significant effects of either intervention on risk perceptions, either for improved varieties in general or for the promoted variety specifically.

A complementary indicator of positive sentiment toward a variety is whether farmers recommend it to others. Peer recommendations are of particular interest given evidence that social learning plays an important role in the diffusion of agricultural technologies and motivates policies that subsidize or temporarily provide seed at no cost (Conley and Udry, 2010; Van Campenhout, 2021). To assess whether the interventions affected peer learning, we asked farmers whether they recommended any improved varieties to others, and, among those familiar with *Bazooka*, whether they recommended *Bazooka* specifically. A substantial share of farmers in the control group (43 percent) reported recommending improved varieties. Receipt of a seed trial pack increased the likelihood of recommending improved varieties by 29 percentage points. Focusing specifically on *Bazooka*, the effect is even larger: farmers who received a seed trial pack were 44 percentage points more likely to recommend the promoted variety.

Finally, we examine farmers' stated intentions regarding future use of improved varieties as an additional indicator of general attitudes. In the control group, 80 percent of farmers report an intention to use improved varieties in the future, while 25 percent report an intention to use Bazooka specifically. We find no treatment effects on stated intentions to use improved varieties in general. However, farmers who received a seed trial pack are 20 percentage points more likely to report an intention to plant Bazooka in the future. This pattern suggests that trial packs primarily shape intentions toward the promoted variety rather than toward improved technologies more broadly, consistent with the observed concentration of adoption responses on Bazooka.

5.2 Trait Perceptions

5.2.1 Perceptions of Consumption Traits

Farmers may be of the opinion that crops grown from local varieties are tastier than crops grown from improved varieties (Pícha, Navrátil, and Švec, 2018; Timu et al., 2014). The cooking demonstrations and tasting sessions were designed to alter these potentially biased perceptions about the consumption qualities of maize grain obtained from improved varieties. Additionally, if after farmers plant the seed trial pack, they process the resulting harvest separately and use it for home consumption, they may revise their beliefs about the consumption traits of these improved varieties as well.

To assess whether our interventions influence farmers' perceptions of the consumption traits of maize from improved seed varieties, we included a dedicated module in the questionnaire that asked farmers to compare maize from local varieties to maize from improved varieties (such as *Longe5*, a popular OPV, or *Bazooka*) across four traits: taste, portion size, appearance, and ease of cooking. For each trait, farmers rated their preference on a 5-point Likert scale, ranging from "local varieties are much better" to "local varieties are much worse." For the purpose of our analysis, we consider improved varieties to be preferred when farmers indicate that local varieties are "somewhat worse" or "much worse" for a given trait.

Results are summarized in Table 6. The overall satisfaction with the consumption traits of improved varieties is already fairly high among control households. Both the seed trial pack intervention and the cooking demonstration and blind tasting sessions significantly increased the proportion of farmers who perceive improved varieties as superior to local varieties according to the summary index, as well as across all four consumption attributes. Interestingly, the results show that the seed trial pack is equally, if not more, effective in changing perceptions in favor of improved seed varieties. This suggests that trial pack interventions may, in some cases, be sufficient to encourage farmers to learn about the consumption traits of new varieties.

5.2.2 Perceptions of Production Traits

The seed trial pack is primarily designed to address preconceived notions farmers may have about the production-related traits of improved varieties. For example, qualitative fieldwork conducted during the study's preparation revealed that some farmers believe that improved maize varieties might offer higher yields but were less resistant to fall armyworm (*Spodoptera frugiperda*), a highly destructive invasive pest that has spread rapidly across East Africa in recent years. Others, having been disappointed in the past, might no longer believe in the yield advantages of improved varieties (Miehe et al., 2025). Providing free seed trial packs can be an effective way to alter these perceptions, as it allows farmers to directly experience the production traits of the new technology on their own fields (Foster and Rosenzweig, 1995).

To test if our interventions alter perceptions of improved maize varieties' production traits, we also include a module in the questionnaire where we ask farmers to compare seed of an improved variety to local seed on five agronomic traits—yield, abiotic stress resistance such as drought or heat resistance, biotic stress resistance such as pests and weed resistance, time to maturity, and seed germination rates.

Results are summarized in Table 7. When farmers compare improved varieties directly to local varieties, the summary index indicates a significant positive impact from the seed trial pack, albeit only at the 10 percent significance level. No significant impact was observed from the cooking demonstration and tasting session, which aligns with expectations as farmers can not observe production traits from that treatment.

Table 5: Knowledge, risk, social learning, and intentions

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Awareness and Attitudes Index	0.26	0.16**	0.07	-0.03	1156
	(0.37)	(0.05)	(0.05)	(0.06)	
Knows *Bazooka* (yes=1)†	0.65	0.29**	0.08	0.17**	1538
	(0.48)	(0.04)	(0.05)	(0.02)	
Number of improved varieties farmer knows	2.21	0.48**	0.50**	-0.16	1532
	(1.43)	(0.16)	(0.16)	(0.24)	
Thinks improved varieties are risky (1=yes)	0.08	0.00	0.01	-0.01	1447
	(0.28)	(0.03)	(0.03)	(0.04)	
Thinks *Bazooka* is risky (1=yes)	0.13	0.03	0.01	-0.05	1207
	(0.33)	(0.05)	(0.06)	(0.08)	
Recommended improved varieties to others (1=yes)	0.43	0.29**	0.02	0.01	1538
	(0.50)	(0.06)	(0.05)	(0.07)	
Recommended *Bazooka* to others (1=yes)	0.34	0.44**	0.11	-0.11	1260
	(0.48)	(0.06)	(0.07)	(0.08)	
Will use improved varieties in the future (1=yes)	0.80	0.05	0.01	0.00	1461
	(0.40)	(0.05)	(0.05)	(0.07)	
Will use *Bazooka* in the future (1=yes)	0.25	0.20**	0.05	-0.07	1503
	(0.43)	(0.06)	(0.05)	(0.09)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). † indicates that baseline outcome was used as control in regression.

Table 6: Impact on perceptions of consumption traits of maize grain obtained from improved seed varieties versus maize grain obtained from local seed varieties

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Consumption Trait Perceptions Index	0.06	0.29**	0.26**	-0.12	1248
	(0.71)	(0.09)	(0.09)	(0.11)	
Improved variety grain tastes better (1=yes)	0.51	0.27**	0.15*	-0.11+	1424
	(0.50)	(0.06)	(0.06)	(0.08)	
Portions obtained from improved variety grain are larger (1=yes)	0.67	0.14**	0.15**	-0.08+	1360
	(0.47)	(0.04)	(0.05)	(0.06)	
Improved variety grain had better appearance (1=yes)	0.77	0.06	0.10*	-0.02	1421
	(0.42)	(0.05)	(0.04)	(0.05)	
Improved variety grain is easier to cook (1=yes)	0.54	0.19**	0.18*	-0.10	1316
	(0.50)	(0.06)	(0.06)	(0.08)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

Table 7: Impact on perceptions of production traits of improved seed varieties versus local seed

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Production Trait Perceptions Index	0.02	0.13^{+}	0.05	-0.04	1273
	(0.60)	(0.08)	(0.07)	(0.10)	
Improved seed yields more (1=yes)	0.90	0.03	0.01	-0.02	1464
	(0.29)	(0.03)	(0.03)	(0.04)	
Improved seed is more restistant to abiotic stresses (1=yes)	0.69	0.10	0.00	-0.06	1336
	(0.47)	(0.05)	(0.06)	(0.08)	
Improved seed is more restistant to biotic stresses (1=yes)	0.52	0.12	0.02	-0.05	1391
	(0.50)	(0.06)	(0.06)	(0.09)	
Improve seed matures faster (1=yes)	0.94	-0.01	-0.02	0.05	1446
	(0.24)	(0.03)	(0.02)	(0.04)	
Improved seed has higher germination rates (1=yes)	0.77	0.09	0.06	-0.02	1439
	(0.42)	(0.05)	(0.04)	(0.06)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

Table 8: Recycling on the randomly selected plot: what the trial pack displaces

	control mean	ITT trial pack	ITT cooking demo	ITT interaction
Used recycled Bazooka	0.05	0.62**	0.02	-0.07
		(0.04)	(0.02)	(0.06)
Used recycled seed of another improved variety	0.16	-0.13**	0.01	0.02
		(0.03)	(0.03)	(0.04)
Used any recycled improved seed	0.21	0.49**	0.03	-0.06
		(0.04)	(0.04)	(0.06)
Used a local variety	0.54	-0.38**	-0.05	0.06
		(0.04)	(0.04)	(0.06)
Used farmer-saved seed (incl. local varieties)	0.74	0.11**	-0.02	-0.00
		(0.03)	(0.04)	(0.05)

Note: All outcomes refer to the randomly selected maize plot and equal one if the condition holds (N=1,495). Column (1) reports pure-control means; columns (2) to (4) report intent-to-treat estimates from the fully interacted specification in Equation 1, estimated without baseline controls. Standard errors clustered at the level of randomization (CR2) in parentheses. Seed is classified as recycled when used for more than one season; the recycling module was administered only for improved varieties, so seed of local varieties is farmer-saved by construction. ** denotes significance at the 1 percent level.

6 Adoption Dynamics Following Seed Trial Packs

A key finding from our study is that farmers who received the seed trial pack were more likely to continue using the distributed variety (*Bazooka*), but mostly did so by reusing seed rather than purchasing fresh seed as recommended. In fact, the turnover it generated ran through recycling rather than purchases of fresh seed. Below we first document what this recycled seed displaced, and then explore possible explanations for the behavior, including limited knowledge about hybrid seed degeneration, perceived profitability of recycling and aversion to market price risks, trust and quality perceptions, and liquidity constraints.

Table 8 decomposes seed recycling on the randomly selected plot. The trial pack raises the use of recycled Bazooka by 62 percentage points. Part of this redirects an existing recycling habit: the use of recycled seed of other improved varieties falls by 13 percentage points. The largest part displaces local varieties, whose use falls by 38 percentage points. The remainder is recycling induced at the margin: measured across all varieties, the probability that a farmer plants saved seed rather than buying fresh rises by 11 percentage points, or, equivalently, the probability of a fresh purchase on the plot falls by the same amount. The much larger increase in recycled improved seed (49 percentage points) is therefore mostly a recomposition of seed farmers were already saving, away from local landraces and toward the promoted hybrid, rather than a net rise in saving. This is consistent with the crowd-out of fresh hybrid and OPV seed reported in Table A11, estimated there at a slightly larger 13 percentage points under a stricter definition that counts only fresh seed from a trusted source. The cooking demonstration moves none of these margins. The trial pack thus mostly redirects the use of farmer-saved seed toward the promoted variety: roughly five sixths of the increase in recycled Bazooka replaces other farmer-saved seed, and about one sixth comes at the expense of purchases in the commercial seed market. Displacing old landraces with a once-recycled hybrid is a real but modest upgrade in the first season: recycled Bazooka out-yields local seed (375 versus 323 kilograms per acre). Because a hybrid loses vigor as it segregates over successive cycles, this first-season advantage is expected to erode toward local-seed levels. Whether the pack raises turnover beyond the first season then depends on whether farmers return to the market once their recycled stock degrades, a dynamic we take up in Section 6.2.

6.1 Limited Knowledge

A first plausible explanation is that farmers may be unaware that hybrid seed, such as *Bazooka*, is not intended for recycling.[12] While baseline knowledge regarding hybrid seed recycling is likely balanced across treatment and control groups due to randomization, the decision to purchase new seed is also influenced by farmers' experience with the yield of the seed they currently have access to and can reuse. Treated farmers observe high yields from the trial pack and may, in the absence of adequate knowledge about hybrid seed degeneration, mistakenly infer that the seed can be reused without penalty. Control farmers, by contrast, are more likely using lower-vigor local varieties, making them more inclined to purchase fresh seed when aiming to improve yields. This mechanism implies that the substitution toward recycling is likely stronger in contexts where baseline knowledge about hybrid seed non-recyclability is low.

We did not formally assess farmers' knowledge of hybrid seed recyclability at baseline. However, during respondent validation sessions held in the form of focus group discussions, farmers generally indicated awareness that hybrid seed is not intended for reuse. They also noted that the yield penalty associated with recycling hybrid seed is gradual, becoming more pronounced with each successive cycle of reuse. This perspective is consistent with our data: farmers who planted fresh *Bazooka* seed achieved an average yield of approximately 535 kilograms of maize per acre. In comparison, farmers using local varieties (excluding recycled *Bazooka*), which have likely gone through successive cycles of reuse, averaged 323 kilograms per acre, while those who recycled *Bazooka* seed just once from the seed trial pack obtained intermediate yields of around 375 kilograms per acre. These comparisons are descriptive rather than causal: seed source is not randomly assigned, and farmers who purchase fresh seed may differ from those who recycle. We use them only to gauge the approximate magnitude of the yield penalty from recycling that farmers face. Read together with the experimental estimates, these magnitudes also account for why the trial pack raised use of the promoted variety by 67 percentage points without raising average yields: most of that use consists of recycled grain, which yields roughly what the seed it displaced—recycled local varieties and the fresh hybrid or OPV seed that treated farmers no longer purchased—would have produced.

6.2 Profitability and Aversion to Market Price Risk

Seed adoption decisions differ from those for standard variable inputs because seed can be reproduced, making replacement choices intertemporal and forward-looking. Classic models of seed replacement emphasize that farmers optimally balance expected yield deterioration of retained seed against the costs and uncertainty associated with purchasing new seed, implying that seed recycling can be a rational response even when improved varieties are available (Heisey and Brennan, 1991).

Figure 2 illustrates estimated profits per acre of maize as a function of the sales price per 100 kg bag, computed by combining the yield estimates reported above with observed price and cost parameters, for three seed types: fresh Bazooka hybrid (535 kg/acre), recycled Bazooka (375 kg/acre), and local seed (323 kg/acre). The calculation assumes that planting one acre with *Bazooka* requires approximately 8 kilograms of seed. At the time of the study, the retail price of *Bazooka* seed was UGX 12,000 per kilogram, resulting in a total seed cost of UGX 96,000 per acre. In contrast, we assume that both local seed and recycled *Bazooka* seed are acquired at negligible cost.

The figure reveals that, across most selling price points, recycling *Bazooka* once yields higher profits than both purchasing fresh *Bazooka* or using local seed. Purchasing fresh *Bazooka* only becomes more profitable than recycling when prices at which maize can be sold exceed approximately UGX 62,000 per 100 kg bag. Given that the median reported sales price at endline in our sample was UGX 60,000 per bag, recycling the harvest from the trial pack

[12] We deliberately chose not to inform farmers about the non-recyclability of the seed in order to preserve the integrity of the experimental design, as including such information would have effectively bundled the trial pack with an informational intervention, thereby complicating causal attribution. From an external validity perspective, this decision also reflects typical market conditions: seed dealers rarely convey this information, and warnings about seed degeneration are often absent from packaging.

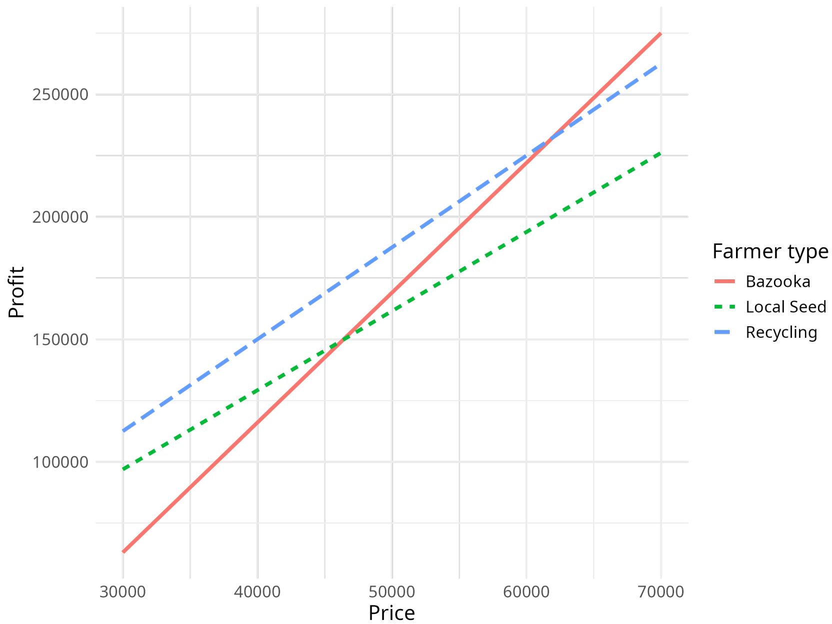

Figure 2: Profits as a function of sales price

for a single season appears to be a profit-maximizing response for the majority of farmers. With its low upfront investment, recycling would also maximize expected utility for a risk averse farmer concerned about any production or price risk that may exist.

Over time, as farmers continue to recycle *Bazooka* seed across multiple seasons, the associated yield (and thus the profit (long-dash blue) line in Figure 2) can be expected to converge toward that of local seed (the short-dash green line). As the yield advantage diminishes, the profitability of recycling declines, making the purchase of fresh hybrid seed increasingly attractive. This has an important, if unobservable, implication for the trial pack. When recycled stock degrades and re-purchase becomes attractive, trial-pack farmers re-enter the market having already learned, trusted, and formed an intention to use the promoted variety (Table 5), so they are plausibly more likely than control farmers to return with improved rather than local seed. If so, the trial pack would raise turnover not as a single pulse but as a recurring cycle of purchase and recycling, lowering the average varietal age over the long run even though the yield gain ebbs between purchases. Our single-season design cannot trace this path, and the magnitude depends on how readily farmers can re-enter the market, which returns us to the frictions discussed below.

It is also important to recognize that the observed high rates of recycling may be season-specific. For example, during the baseline period, maize prices ranged from UGX 100,000 to UGX 120,000 per 100 kg bag—well above the threshold at which purchasing fresh *Bazooka* becomes the dominant strategy. However, since farmers must make input decisions before prices are known, risk-averse farmers may prefer to avoid the upfront investment cost of fresh seed and opt to recycle instead. This also implies that risk aversion is an important cross-cutting factor that amplifies the attractiveness of recycling.

6.3 Quality Issues

Another possible explanation for the observed differences in recycling rates across treatment groups is that farmers may place greater trust in seed provided by a research organization compared to seed sourced from local agro-dealers. This interpretation is consistent with existing evidence documenting low levels of trust in Uganda's agricultural input market (Bold et al., 2017; Barriga and Fiala, 2020; Miehe et al., 2023). To explore this hypothesis, we compare farmer ratings of *Bazooka* seed across different sourcing channels. In addition to eliciting comparative assessments of improved versus local seed varieties on a range of production traits (see Section 5.2.2), we asked farmers to rate the actual seed used on their plots with respect to specific traits, as well as to provide an overall satisfaction score.

The results, summarized in Figure 3, support the hypothesis that seed quality uncertainty also plays a role. On most production traits, *Bazooka* seed obtained through the seed trial pack receives the highest ratings. For instance, nearly 50 percent of farmers assign it the top score for yield performance, and it is also highly rated for its germination rate. These positive assessments are reflected in high overall satisfaction: approximately 80 percent of farmers report being very satisfied with the seed received through the trial pack. Satisfaction declines somewhat among farmers who recycled *Bazooka* from the trial pack, but even recycled seed is generally rated more favorably than fresh *Bazooka* seed purchased from agro-dealers.

6.4 Liquidity Constraints

Cash or credit constraints may also induce recycling. If a farmer wants to plant *Bazooka*, treatment farmers can fall back on recycling as a cost-minimizing strategy, while control farmers likely must purchase seeds. As a result, liquidity constraints may increase the likelihood of recycling among treatment farmers, while selection into *Bazooka* use among control farmers means that only those able to afford fresh seed will use *Bazooka*, resulting in a higher observed rate of fresh seed use among control farmers.

We explore this by looking at the correlation between wealth (proxied by consumption expenditure per capita at baseline) and the likelihood that farmers recycle seed from the seed trial pack (as opposed to buying fresh *Bazooka* seed) in the group that received a seed trial pack (Figure 4). Recycling of *Bazooka* seed is common across all wealth groups. In the lowest quintile, more than 90 percent of farmers recycled, compared to roughly 84–86 percent in the middle quintiles and 77 percent in the fourth quintile. Overall, the data suggest only modest variation in recycling behavior by wealth, with slightly lower recycling rates observed among relatively wealthier farmers.

7 Cost-effectiveness and Scaling

The preceding sections establish that both instruments raise use of the promoted variety, but through different channels and with different consequences for productivity. For a policymaker choosing between them, what matters is not only whether each works but what each costs to deliver. Because we find no evidence that the two instruments are complements (the interaction terms in our results tables are small and imprecisely estimated, although the trial was powered to detect only large interaction effects), little appears to be gained by deploying both, and we treat the choice as one between single instruments; treating the instruments as separable is also what lets us cost them independently. We therefore combine the experimental effects with the actual intervention budgets to compare the two instruments on a cost-effectiveness basis. Because the seed trial packs in our study were distributed during the baseline visit, their realized marginal delivery cost was close to zero. To keep the comparison fair to the cooking demonstration, which was a stand-alone field operation that carries its full delivery cost, we instead cost a stand-alone seed-pack distribution program, using the same unit costs for transport, per diems, and mobilization drawn from the consumption-intervention budget.

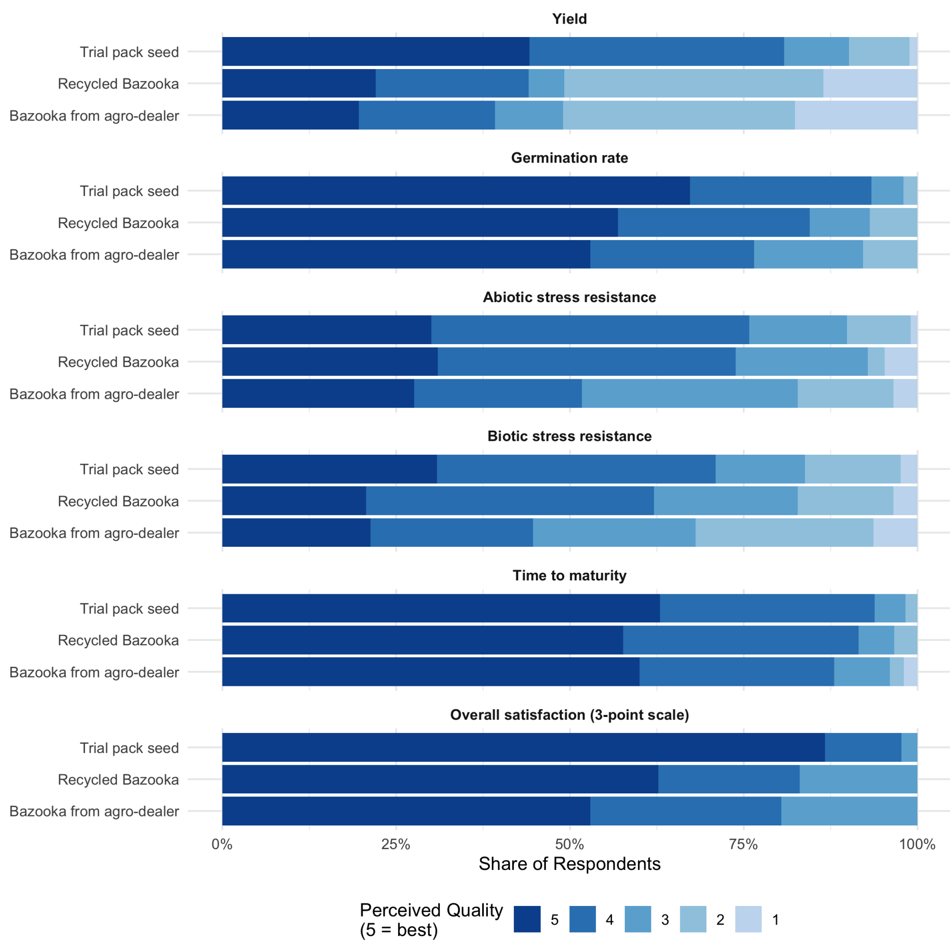

Figure 3: Perceived quality of *Bazooka* seed by source, among trial-pack recipients who grew *Bazooka* on the randomly selected plot

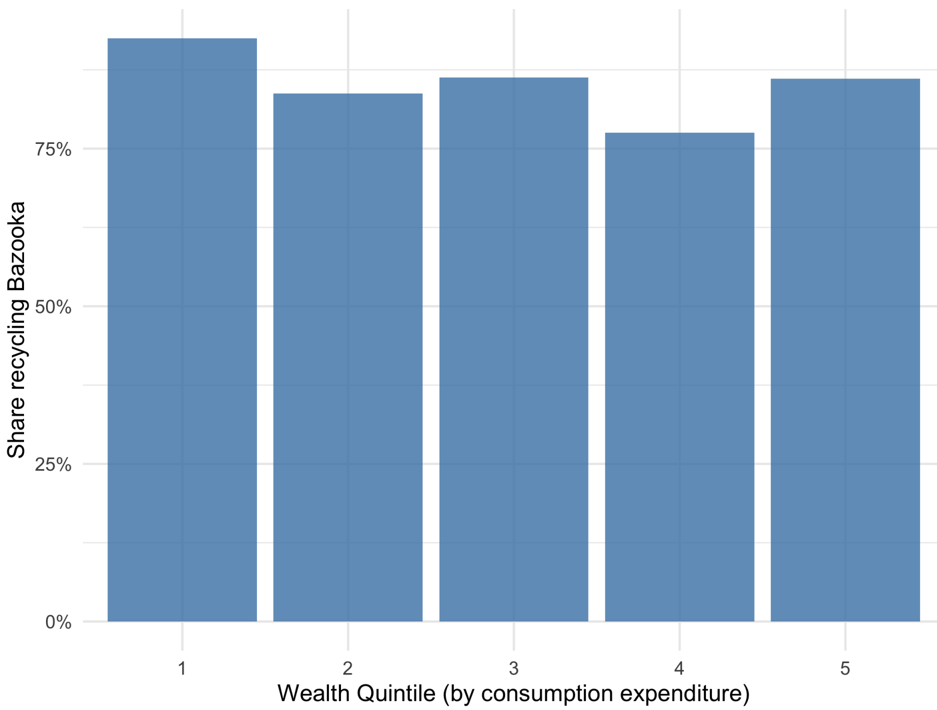

Figure 4: Share of trial-pack recipients growing *Bazooka* who recycled rather than purchased fresh seed, by quintile of baseline consumption expenditure (extreme expenditure values trimmed)

Delivering a seed trial pack costs approximately UGX 27,000 per farmer: UGX 12,000 for one kilogram of seed, plus about UGX 15,000 for stand-alone distribution.[13] A cooking demonstration costs substantially more, at about UGX 75,000 per farmer at full economic cost, or roughly UGX 64,000 once research-specific overhead such as study supervision and enumerator insurance is removed. The difference reflects the demonstration's heavier logistics: each session requires a team to travel to the village, cook, and run a blind tasting, and each attendee receives a two-kilogram flour sample to take home.

These figures capture costs to the implementer; the two instruments also place very different demands on farmers themselves. Attending a cooking demonstration costs a household roughly half a day of time, often for two members, and delivers its information immediately. The trial pack instead requires the farmer to commit a plot and a full season of land and labor before any information is realized. Because the farmer keeps the harvest, the net monetary cost of participating is likely negative—the pack is a transfer as much as a message—so the relevant farmer-side costs are the delay and the land and labor commitment rather than cash outlays. This asymmetry also means the demonstration can reach farmers for whom on-farm experimentation is not feasible, for instance because land is scarce, whereas the trial pack presupposes the capacity and willingness to experiment.

Combining these costs with the intent-to-treat effects on adoption reported in Table 3 shows that the most cost-effective instrument depends on the policy objective. If the objective is varietal turnover, that is, getting an improved variety into the ground regardless of seed source, the seed trial pack costs about UGX 40,000 per additional adopter (UGX 27,000 divided by a 0.67 effect on use of the promoted variety), against roughly UGX 715,000 for the cooking demonstration (UGX 64,000 divided by a 0.09 effect). The trial pack is more than an order of magnitude more cost-effective, and the ranking is robust to how distribution is costed: even if the pack were charged the demonstration's entire stand-alone delivery cost, it would still cost about UGX 112,000 per additional adopter, roughly six times less than the demonstration. If, instead, the objective is to build demand in the commercial seed market, that is, purchases of fresh certified seed, the ranking reverses. The trial pack's estimated effect on fresh-seed purchases is indistinguishable from zero, because the turnover it generates runs through recycling, so spending on trial packs does not buy fresh-seed adoption through this channel and no meaningful cost per fresh-seed adopter can be computed. The cooking demonstration is the only instrument for which we find evidence, albeit suggestive, of increased fresh-seed purchases; the implied figure of about UGX 1.3 million per additional fresh adopter inherits the imprecision of that estimate and is best read as an order of magnitude, though it comes alongside suggestive yield gains.

An important qualification concerns the time horizon. We observe one planting decision after each instrument, and the fresh-seed purchases induced by the demonstration are first acquisitions, made by farmers who hold no grain of the promoted variety to replant. Once these farmers harvest, they face the same recycling incentives documented in Section 6, and may recycle rather than buy again. On the evidence we have, the demonstration therefore shifts the entry margin into the certified-seed market; its effect on repeat purchases lies outside our observation window. The same horizon logic applies in reverse to the trial pack: because the yield advantage of recycled hybrid grain erodes with each cycle of reuse, the large cohort of recycling farmers the pack creates may itself convert into fresh-seed demand in later seasons. Within the window we observe, however, the pack substitutes for commercial purchases while the demonstration adds to them.

The two instruments also have very different cost structures, with implications for scaling. The seed trial pack is almost entirely variable cost: the seed is a fixed quantity per farmer that cannot be shared across recipients. The cooking demonstration, by contrast, combines modest fixed costs, namely reusable equipment and one-off training together amounting to about UGX 10 million, with a large per-session cost for transport, staff, and cooking that is shared across everyone who attends. As a result, the per-farmer cost of the two instruments responds differently to scale. Spreading the program across more villages does not change their relative ranking, because the

[13] All costs are in 2023 Ugandan shillings (UGX). At the average 2023 exchange rate of roughly UGX 3,700 per US dollar, the trial pack costs about US$7 per farmer and the cooking demonstration about US$17 to US$20.

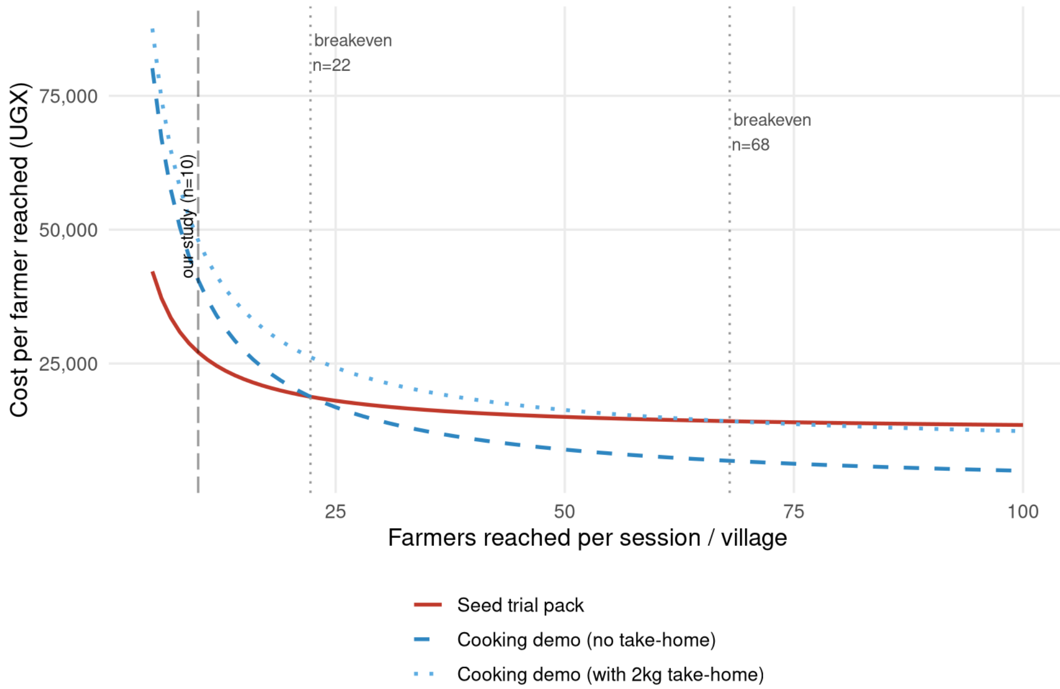

Figure 5: Cost per farmer reached, by number of farmers per session

demonstration's per-village transport and staff costs are variable rather than fixed and do not fall as the number of villages rises. What does shift the ranking is the number of farmers reached per session: because the demonstration's dominant costs are shared across attendees while the pack's dominant cost is borne per farmer, inviting more farmers to each demonstration lowers its per-farmer cost while leaving the pack's essentially unchanged.

Figure 5 illustrates this, plotting the cost per farmer reached against the number of farmers per session. In our study, each session reached about 10 farmers, and at that scale the trial pack is cheaper. But the demonstration's cost per farmer falls steeply with attendance, crossing below the trial pack at about 22 farmers per session for a stripped-down demonstration without an individual take-home sample, and at about 68 farmers if the two-kilogram take-home flour is retained.[14] A village-wide demonstration of 30 to 50 farmers, without per-head take-home flour, would therefore reach farmers more cheaply than distributing trial packs.

Two caveats temper this conclusion. First, the crossover is expressed in cost per farmer reached, not cost per adopter. Our adoption effect for the demonstration was estimated at roughly 10 attendees per session; concentrating fifty farmers in a single demonstration may dilute its effectiveness, as attendees further from the cooking and tasting learn less. The cost-effectiveness crossover therefore requires the demonstration's per-attendee effect to survive larger audiences, which we cannot observe in our data. Second, the demonstration's effect on adoption is itself imprecisely estimated, so the cost-per-adopter figures for that instrument carry wide confidence intervals. With

[14]The breakeven equates per-farmer costs, $S/n + v = c + D/n$, where S (about UGX 396,000) is the per-session cost of a demonstration shared across attendees, D (about UGX 151,000) is the per-village cost of distributing packs, c (UGX 12,000) is the seed cost per farmer, and v is the per-attendee variable cost of a demonstration (about UGX 1,000 for the tasting alone, about UGX 8,400 including the two-kilogram take-home sample). Solving for n gives $(S-D)/(c-v)$, or about 22 and 68 attendees, respectively.

these qualifications, the practical implication is clear: trial packs are the cheaper instrument for raising varietal turnover at the scale at which extension is typically delivered, while cooking demonstrations become cost-competitive only when delivered to large groups, and are valuable mainly where the objective is to stimulate initial fresh-seed purchases rather than turnover alone.

8 Conclusion

In this study, we evaluate two commonly used extension strategies aimed at promoting the adoption of an improved maize variety among smallholder farmers in eastern Uganda: seed trial packs emphasizing production attributes and cooking demonstrations emphasizing consumption attributes. Using a cluster-randomized field experiment with a 2×2 factorial design, we show that both interventions shape farmers' perceptions and use of the promoted variety, but through distinct pathways and with different implications for productivity.

The seed trial pack substantially increased farmers' awareness and appreciation of both production- and consumption-related attributes and led to a marked increase in subsequent use of the promoted variety. However, this adoption response was driven primarily by the recycling of harvested grain as seed. For hybrid varieties like the variety we used in the study, such recycling entails a yield penalty, and consistent with this, we find no evidence of yield gains despite increased use. In contrast, cooking demonstrations and tasting sessions primarily influenced perceptions of consumption attributes. While their effects on adoption were more modest, farmers exposed to this intervention were more likely to use fresh seed, and we find suggestive evidence of associated yield gains.

Taken together, these findings highlight the importance of aligning extension strategies with the technological characteristics of the promoted seed. Trial-based extension appears well suited to technologies that are compatible with farmer-led reproduction, such as open-pollinated varieties. Applied to a hybrid, however, the trial pack induces the very behavior that erodes the variety's advantage: recycling. The gain over the displaced seed is real in the first season, but it decays as the hybrid segregates over successive cycles. Because the pack also durably raises awareness of, trust in, and intention to use the variety (Table 5), trial-pack farmers may re-enter the market with improved seed once their recycled stock degrades, so the turnover it sets in motion is plausibly recurring rather than one-off, though our single-season design cannot trace its path. This holds even though most of the recycling displaces farmer-saved local seed rather than market purchases: of the 62 percentage point increase in recycled Bazooka use, 38 points come out of local varieties, 13 out of recycled seed of other improved varieties, and only 11 out of fresh purchases. The pack moves farmers off old landraces and onto a recent hybrid; keeping that gain improved over time, however, requires periodic re-purchase, and thus a seed market farmers can return to.

For policymakers, the practical message is that the right instrument depends on the objective. If the aim is a quick, low-cost rise in the share of fields under an improved variety, seed trial packs dominate: they are roughly an order of magnitude cheaper per additional adopter than cooking demonstrations. If instead the aim is to build a self-sustaining commercial seed market, trial packs are the wrong tool, because the turnover they generate runs through recycling and leaves purchases of fresh certified seed unchanged. Cooking demonstrations are the only instrument we find to move that margin, but the effect is suggestive, costly per adopter, and—because attendees buy fresh seed partly for lack of anything to recycle—likely confined to first purchases. Sustained commercial demand will therefore depend less on either extension instrument than on easing the frictions that make recycling the rational choice: seed prices relative to grain prices, unreliable input quality, and limited liquidity. Combining the two instruments adds little: we find no evidence of complementarity between them, so a budget is better spent on the single best-matched instrument than split across both. And because the cost of a cooking demonstration is largely incurred per session and shared across those who attend, its cost-effectiveness improves markedly when it is delivered to large groups rather than the small sessions we study.

More broadly, our results underscore that observed adoption patterns reflect rational responses to economic and structural constraints rather than simple information failures. For the average farmer in our sample, recycling seed appears to be a reasonable choice given modest yield penalties, high seed costs relative to grain prices, unreliable input quality, and limited liquidity. These findings suggest that efforts to promote hybrid seed adoption may need to be complemented by interventions that relax liquidity constraints, improve access to affordable high-quality seed, or strengthen output market incentives that reward productivity gains.

Finally, while consumption traits clearly matter for varietal choice, our results point to a division of labor between the two instruments. For learning about consumption attributes, demand-creation events are not strictly necessary: farmers who received only a trial pack updated their perceptions of consumption traits about as much as demonstration attendees, presumably by cooking and tasting their own harvest. The distinctive value of cooking demonstrations lies elsewhere—in moving farmers to a first purchase of fresh seed, the margin that trial packs leave untouched. Future research should further explore how different modes of experiential learning interact with technology characteristics and market conditions to shape adoption dynamics. Understanding these interactions is crucial for designing extension strategies that promote not only uptake, but also sustained, productivity-enhancing use of improved crop technologies.

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Online Appendix

A1 Pooled models

In this Online Appendix, we provide results for the same set of outcomes and in similar tables as in the main manuscript, but now estimated using an alternative empirical specification. Note that in these pooled specifications the comparison group for each treatment consists of all households not assigned to that treatment, so reported control means can differ slightly from those in the main tables. Factorial designs are commonly employed to evaluate multiple treatments within a single experiment. While our design is powered for a complete set of interactions (as in equation 1), our pre-analysis plan also states that we may wish to increase statistical power by pooling observations across the orthogonal treatments if we find that a treatment effect is smaller than the minimal detectable effect size assumed during power calculations. However, pooling treatment cells to enhance statistical power can lead to biased estimates of the main treatment effects if there is an interaction between the treatments (Muralidharan, Romero, and Wüthrich, 2023). To mitigate this, we will treat the orthogonal treatment as a covariate we want to adjust for and interact the treatment variable with the demeaned orthogonal treatment (Lin, 2013):

$$Y_{ij} = \alpha + \beta_M T_i^M + \beta_O \left(T_i^O - \bar{T^O} \right) + \beta_I T_i^M \left(T_i^O - \bar{T^O} \right) + \delta Y_{0ij}^B + \varepsilon_{ij} \tag{2}$$

where now T_i^M is a dummy for the main treatment (the seed trial pack or the cooking demonstration respectively) and T_i^O is a dummy for the orthogonal treatment (which enters in deviations from its means).

It is important to note that the estimand of the respective treatments in Equation 2 changes. For instance, while the estimand for the seed trial pack treatment in Equation 1 is the effect of the treatment in and of itself, the estimand in Equation 2 is a weighted average of the estimand of the seed trial pack treatment of Equation 1 and the effect of the seed trial pack in the presence of the cooking demonstration and testing treatment.

Tables A1 through A8 report results for the pooled models (Equation 2) and correspond to the fully interacted models presented in Tables 1 through 7. In these tables, the second column corresponds to β_M in Equation 2 for the seed trial pack treatment (in which case the cooking demonstration and tasting treatment is considered the orthogonal treatment and controlled for). In the fourth column, this is reversed, showing β_M in Equation 2 for the cooking demonstration and tasting treatment (with the seed trial pack treatment now considered the orthogonal treatment and controlled for). The first and the third columns are control group means for the respective treatments.

Table A1: Baseline balance (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo
Age of household head (in years)	48.78	-0.31	48.94	-0.65
	(13.79)	(0.85)	(13.57)	(0.85)
Household head has finished primary education (1-yes)	0.50	0.02	0.51	0.00
	(0.50)	(0.03)	(0.50)	(0.03)
Gender of household head (1=male)	0.82	-0.03	0.82	-0.04^{+}
	(0.39)	(0.03)	(0.38)	(0.03)
Household size	8.24	-0.02	8.14	-0.01
	(3.70)	(0.03)	(3.57)	(0.03)
Distance of homestead to nearest agro-input shop (km)	4.18	-0.01	4.04	0.04
	(3.74)	(0.10)	(3.61)	(0.10)
Has used the promoted seed (*Bazooka*) on any plot in the last season (1=yes)	0.09	0.01	0.10	-0.01
	(0.29)	(0.02)	(0.30)	(0.02)
Has used quality maize seed on any plot in last season (1=yes)	0.42	-0.04	0.41	-0.02
	(0.49)	(0.03)	(0.49)	(0.03)
Seed on random plot was obtained from formal seed source (1=yes)	0.32	-0.04	0.28	0.02
	(0.47)	(0.03)	(0.45)	(0.03)
Used seed that is recycled more than 3 seasons on randomly selected plot (1=yes)	0.53	0.02	0.53	0.03
	(0.50)	(0.04)	(0.50)	(0.04)
Maize yields on a randomly chosen plot in last season (kg/acres)	394.85	-0.06	379.59	0.00
	(301.60)	(0.06)	(287.58)	(0.06)

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels. Household size, distance to nearest agro-input shop, and productivity is transformed using the inverse hyperbolic sine transformation (means and standard deviations are in levels).

Table A2: Adoption (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Adoption Index	0.05	0.33**	0.18	0.05	1495
	(0.84)	(0.05)	(0.75)	(0.05)	
Has used *Bazooka* on any plot in last season in last season† (1=yes)	0.14	0.63**	0.44	0.03	1495
	(0.35)	(0.03)	(0.50)	(0.03)	
Number of plots planted with improved varieties	0.18	0.78**	0.56	0.01	1495
	(0.49)	(0.04)	(0.74)	(0.04)	
Number of plots with improved varieties as share of total number of plots	0.13	0.60**	0.42	0.01	1495
	(0.33)	(0.03)	(0.49)	(0.03)	
Area planted with improved varieties (acres)	0.17	0.67**	0.50	0.01	1495
	(0.49)	(0.06)	(0.85)	(0.06)	
Area planted with improved varieties as a share of total maize cultivation area	0.13	0.60**	0.42	0.01	1495
	(0.33)	(0.03)	(0.49)	(0.03)	
Has used fresh *Bazooka* on any plot in last season in last season† (1=yes)	0.08	0.02	0.06	0.05**	1495
	(0.27)	(0.02)	(0.25)	(0.02)	
Has used seed of an improved variety on any plot in last season† (1=yes)	0.31	-0.13**	0.23	0.04^{+}	1488
	(0.46)	(0.03)	(0.42)	(0.02)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). † indicates that baseline outcome was controlled for in the regression.

Table A3: Adoption on random plot (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Adoption Index	0.05	0.22**	0.10	0.08*	1345
	(0.70)	(0.04)	(0.64)	(0.04)	
Has used *Bazooka* on randomly selected plot in last season (yes=1)	0.13	0.60**	0.42	0.03	1495
	(0.34)	(0.03)	(0.49)	(0.03)	
Quantity of improved variety used on randomly selected plot (kg)	0.74	3.01**	2.11	0.05	1437
	(2.44)	(0.25)	(3.44)	(0.25)	
Quantity of improved variety used on randomly selected plot (kg/acre)	0.79	4.09**	2.72	0.00	1427
	(2.38)	(0.25)	(4.06)	(0.25)	
Maize production (log)	5.43	-0.07	5.33	0.11^{+}	1410
	(1.01)	(0.06)	(0.98)	(0.06)	
Maize productivity (log)	5.66	0.01	5.59	0.13*	1375
	(0.87)	(0.05)	(0.88)	(0.05)	
Has used fresh *Bazooka* on randomly selected plot in last season (yes=1)	0.07	0.01	0.06	0.04*	1495
	(0.25)	(0.01)	(0.23)	(0.01)	
Has used improved variety on randomly selected plot in last season (yes=1)	0.29	-0.12**	0.21	0.03	1495
	(0.45)	(0.02)	(0.41)	(0.02)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). All regressions include baseline outcome as control.

Table A4: Use of harvest (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Use Index	-0.01	0.06	-0.01	0.03	1302
	(0.71)	(0.05)	(0.74)	(0.05)	
Share of maize consumed (%)	75.19	0.63	75.19	-0.72	1449
	(29.93)	(2.11)	(30.32)	(2.10)	
Share of maize sold (%)	21.42	-0.74	21.42	-0.31	1495
	(29.11)	(2.08)	(30.05)	(2.08)	
Share of maize kept for seed (%)	4.74	-0.83	4.74	-0.48	1321
	(8.02)	(0.61)	(8.36)	(0.61)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

Table A5: Impact on women co-head involvement (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Women's Empowerment Index	-0.09	0.12^{+}	0.03	-0.11	1098
	(1.03)	(0.07)	(0.90)	(0.07)	
Woman involved in decision what to plant (1=yes)	0.82	0.04	0.86	-0.03	1098
	(0.38)	(0.03)	(0.35)	(0.03)	
Women involved in what to do with harvest (1=yes)	0.83	0.05	0.88	-0.04	1098
	(0.37)	(0.03)	(0.33)	(0.03)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). All regressions include baseline outcome as control.

Table A6: Knowledge, risk, social learning, and intentions (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Awareness and Attitudes Index	0.30	0.14**	0.36	0.05+	1156
	(0.36)	(0.03)	(0.38)	(0.03)	
Knows *Bazooka* (yes=1)†	0.69	0.26**	0.79	0.06+	1538
	(0.46)	(0.03)	(0.41)	(0.03)	
Number of improved varieties farmer knows	2.46	0.40**	2.45	0.42**	1532
	(1.48)	(0.12)	(1.46)	(0.12)	
Thinks improved varieties are risky (1=yes)	0.09	0.00	0.08	0.00	1447
	(0.28)	(0.02)	(0.28)	(0.02)	
Thinks *Bazooka* is risky (1=yes)	0.13	0.01	0.14	-0.02	1207
	(0.34)	(0.04)	(0.35)	(0.04)	
Recommended improved varieties to others (1=yes)	0.44	0.29**	0.58	0.02	1538
	(0.50)	(0.04)	(0.49)	(0.04)	
Recommended *Bazooka* to others (1=yes)	0.40	0.38**	0.60	0.06	1260
	(0.49)	(0.04)	(0.49)	(0.04)	
Will use improved varieties in the future (1=yes)	0.80	0.05	0.82	0.01	1461
	(0.40)	(0.03)	(0.38)	(0.03)	
Will use *Bazooka* in the future (1=yes)	0.28	0.16**	0.35	0.02	1503
	(0.45)	(0.04)	(0.48)	(0.04)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). † indicates that baseline outcome was used as control in regression.

Table A7: Impact on perceptions of consumption traits of maize grain obtained from improved seed varieties versus maize grain obtained from local seed varieties (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Consumption Trait Perception Index	0.20	0.23**	0.23	0.20**	1248
	(0.65)	(0.05)	(0.66)	(0.05)	
Improved variety grain tastes better (1=yes)	0.59	0.21**	0.65	0.09**	1424
	(0.49)	(0.04)	(0.48)	(0.04)	
Portions obtained from improved variety grain are larger (1=yes)	0.75	0.10**	0.75	0.10**	1360
	(0.43)	(0.03)	(0.43)	(0.03)	
Improved variety grain had better appearance (1=yes)	0.83	0.05*	0.80	0.09**	1421
	(0.38)	(0.03)	(0.40)	(0.03)	
Improved variety grain is easier to cook (1=yes)	0.64	0.14**	0.65	0.13**	1316
	(0.48)	(0.04)	(0.48)	(0.04)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

Table A8: Impact on perceptions of production traits of improved seed varieties versus local seed (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Production Trait Perceptions Index	0.04	0.11*	0.09	0.03	1273
	(0.60)	(0.05)	(0.58)	(0.05)	
Improved seed yields more (1=yes)	0.91	0.03	0.92	0.00	1464
	(0.29)	(0.02)	(0.27)	(0.02)	
Improved seed is more restistant to abiotic stresses (1=yes)	0.69	0.06	0.73	-0.03	1336
	(0.46)	(0.04)	(0.44)	(0.04)	
Improved seed is more restistant to biotic stresses (1=yes)	0.53	0.09	0.59	-0.01	1391
	(0.50)	(0.04)	(0.49)	(0.04)	
Improve seed matures faster (1=yes)	0.93	0.01	0.93	0.00	1446
	(0.26)	(0.02)	(0.25)	(0.02)	
Improved seed has higher germination rates (1=yes)	0.80	0.08^{+}	0.82	0.05	1439
	(0.40)	(0.03)	(0.39)	(0.03)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

A2 Well-being and Food Security

This section provides more details on the impact of seed trial packs and the cooking and tasting session on food security and welfare indicators. Results for the fully interacted models are in Table A9, while results for the pooled models are in Table A10.

Judging by the index values, we indeed find no effects from the treatments on overall welfare and food security. The effects on individual indicators broadly tell the same story, showing little in the way of welfare gains. The first two indicators assess subjective measures of relative well-being, both within the village and over time. In particular, we asked if the respondent felt that their household was better off, about the same, or worse off in terms of income and consumption than the average households in the community and create an indicator variable which is equal to one if the respondent answered that they were better off and zero otherwise. Similarly, we asked if the respondent felt that their household was better off, about the same, or worse off in terms of income and consumption than six months earlier and construct an analogous indicator. Approximately 35 percent of farm households report that they perceive themselves as better off compared to the average within their village. There is no significant change in this perception following either the seed trial pack nor the cooking demonstration and tasting session. Additionally, 38 percent of households believe they are better off than they were six months ago—approximately one agricultural season prior. Notably, there is some indication that farmers in the seed trial pack group are more likely to feel that their well-being improved over time compared to those who did not receive a seed trial pack (though the effect does not survive p-value adjustments for multiple hypothesis testing). This may be due to the perceived benefits from planting the seed trial pack, either through real or perceived harvest increases (despite yields not showing a significant increase on average, as detailed in Table A11).

Next, we use two questions from the The Household Food Insecurity Access Scale (HFIAS), including one that aligns with the "food preference" dimension, which assesses whether people had to compromise on their preferred diet due to financial or food constraints, and one that captures "reductions in food quantity", a more severe form of food insecurity where people eat smaller portions or skip meals due to lack of resources (Coates, Swindale, and Bilinsky, 2007).[15] Table A9 shows that 52 percent of households in the control group indicate that in the past month they were always able to eat what they wanted, whereas 61 percent of control households indicate that in the past month they always had the desired quantities of food they wanted. We do not find any impact of the interventions on these outcomes.

Finally, we approximate consumption expenditure by asking farmers to report how much money was spent in the week prior to the survey on 14 of the most consumed items in the area (maize, sorghum, millet, rice, cassava, sweet potatoes, beans, groundnuts, fruits, vegetables, sugar, cooking oil, soap, and airtime) and dividing this by the household size. On average, a household in the control group spent about 12234 per capita, which at the time of the survey corresponds to about US$3. We also do not find that our treatments affected per capita consumption expenditure.

[15] In our analysis, as we want to focus on positive outcomes, we take the inverse and measure food security instead of insecurity.

Table A9: Welfare and food security

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Welfare and Food Security Index	0.01	0.00	-0.08	0.13	1471
	(0.58)	(0.06)	(0.06)	(0.09)	
HH is better off than average of village? (1=yes)	0.35	0.00	-0.02	0.05	1504
	(0.48)	(0.05)	(0.05)	(0.07)	
HH is better off than 6 months ago? (1=yes)	0.38	0.10	-0.05	0.06	1531
	(0.49)	(0.05)	(0.05)	(0.07)	
HH always eat what they want (1=yes)	0.52	-0.07	-0.07	0.21	1538
	(0.50)	(0.07)	(0.07)	(0.10)	
HH can always eat quantity needed (1=yes)	0.61	0.01	-0.10	0.13	1538
	(0.49)	(0.05)	(0.06)	(0.08)	
Consumption expenditure (*1000 UGX/week/capita)	12234	169	-59	-525	1507
	(8603)	(792)	(908)	(1176)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

Table A10: Welfare and food security (pooled)

	control mean	ITT trial pack	control mean	ITT cooking demo	nobs
Welfare and Food Security Index	0.01	0.06	0.01	-0.01	1471
	(0.59)	(0.04)	(0.57)	(0.04)	
HH is better off than average of village (1=yes)	0.35	0.02	0.35	0.00	1504
	(0.47)	(0.03)	(0.48)	(0.03)	
HH is better off than 6 months ago (1=yes)	0.38	0.12**	0.38	-0.02	1531
	(0.48)	(0.03)	(0.50)	(0.03)	
HH can always eat what they want (1=yes)	0.52	0.03	0.52	0.03	1538
	(0.50)	(0.05)	(0.50)	(0.05)	
HH can always eat quantity needed (1=yes)	0.61	0.08	0.61	-0.04	1538
	(0.50)	(0.04)	(0.49)	(0.04)	
Consumption expenditure (*1000 UGX/week)	12233.87	-93.00	12233.87	-320.31	1507
	(8494.61)	(588.00)	(8066.98)	(585.79)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006).

A3 Additional Tables

A4 Heterogeneity by Prior Exposure to Improved Seed

Table A13 reports treatment effects separately for farmers who had and had not used an improved maize variety, a fresh or recycled OPV or hybrid, in the season before the intervention (about 40 percent of the sample). Estimates come from the pooled specification in which each treatment is interacted with the baseline-use indicator, with standard errors clustered at the village level. These subgroup analyses were not pre-specified and the study is powered for average effects, so they should be read as exploratory.

Table A11: Adoption on random plot

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Adoption Index	-0.01	0.22**	0.09^{+}	-0.01	1345
	(0.71)	(0.05)	(0.05)	(0.07)	
Has used *Bazooka* on randomly selected plot in last season (yes=1)	0.10	0.64**	0.07	-0.08	1495
	(0.29)	(0.04)	(0.03)	(0.06)	
Quantity of *Bazooka* used on randomly selected plot (kg)	0.55	3.18**	0.22	-0.34	1437
	(1.88)	(0.36)	(0.23)	(0.49)	
Quantity of *Bazooka* used on randomly selected plot (kg/acre)	0.64	4.28**	0.19	-0.38	1427
	(2.15)	(0.32)	(0.22)	(0.50)	
Maize production (log)	5.39	-0.09	0.09	0.04	1410
	(1.02)	(0.08)	(0.09)	(0.12)	
Maize productivity (log)	5.61	-0.01	0.11	0.04	1375
	(0.91)	(0.08)	(0.07)	(0.11)	
Has used fresh *Bazooka* on randomly selected plot in last season (yes=1)	0.04	0.02	0.05*	-0.02	1495
	(0.20)	(0.02)	(0.02)	(0.03)	
Has used fresh hybrid or OPV on randomly selected plot in last season (yes=1)	0.27	-0.13**	0.02	0.02	1495
	(0.45)	(0.03)	(0.03)	(0.05)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). All regressions include baseline outcome as control.

Table A12: Impact on women co-head involvement

	control mean	ITT trial pack	ITT cooking demo	ITT interaction	nobs
Women's Empowerment Index	0.00	0.05	-0.18	0.15	1098
	(0.95)	(0.10)	(0.11)	(0.14)	
Woman involved in decision what to plant (1=yes)	0.85	0.01	-0.06	0.05	1098
	(0.36)	(0.04)	(0.04)	(0.05)	
Women involved in what to do with harvest (1=yes)	0.87	0.02	-0.07	0.05	1098
	(0.34)	(0.04)	(0.04)	(0.05)	

Note: Column (1) reports control group means at baseline (and standard deviations below); column (2) reports the estimate of the intent-to-treat effect of the seed trial packs; column (3) reports the estimate of the intent-to-treat effect of the cooking and tasting demonstrations; column (4) is the interaction effect. Standard errors are below the estimates and are clustered at the level of randomization; **, *, and + denote significance at the 1, 5, and 10% levels based on sharpened q-values (Benjamini, Krieger, and Yekutieli, 2006). All regressions include baseline outcome as control.

Table A13: Treatment effects by prior use of improved seed

Instrument	Outcome	No prior use	Prior use	Difference
Trial pack	Adoption index	0.48**	0.13+	-0.35**
		(0.05)	(0.07)	(0.09)
Trial pack	Used any Bazooka	0.72**	0.50**	-0.22**
		(0.03)	(0.05)	(0.05)
Trial pack	Used fresh Bazooka	0.02	0.01	-0.02
		(0.02)	(0.03)	(0.03)
Cooking demo	Adoption index	0.05	0.08	0.03
		(0.06)	(0.08)	(0.09)
Cooking demo	Used any Bazooka	0.03	0.04	0.02
		(0.03)	(0.05)	(0.05)
Cooking demo	Used fresh Bazooka	0.03*	0.07*	0.04
		(0.02)	(0.03)	(0.03)

Note: Intent-to-treat effects from the pooled specification, each treatment interacted with an indicator for baseline use of an improved variety (about 40 percent of the sample). The first two columns give the implied effect for farmers without and with prior use; Difference is the interaction term. Standard errors clustered at the village level (CR2) in parentheses. Subgroup analyses are exploratory and were not pre-specified. ** $p<0.01$, * $p<0.05$, + $p<0.10$.